SNARGs for NP and Non-Signaling PCPs, Revisited

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Abstract

We revisit the question of whether it is possible to build succinct non-interactive arguments (SNARGs) for all of NP under standard assumptions using non-signaling probabilistically checkable proofs [Kalai-Raz-Rothblum, STOC' 14]. In particular, we observe that using *exponential-length* PCPs appears to circumvent all of the existing barriers.

For our main result, we give a candidate non-adaptive SNARG for NP and prove its soundness under:

1. the learning with errors assumption (or other standard assumptions such as bilinear maps), and
2. a mathematical conjecture about multivariate polynomials over the reals.

In more detail, our conjecture is an upper bound on the minimum total coefficient size of Nullstellensatz proofs (Potechin-Zhang, ICALP 2024) of membership in a concrete polynomial ideal. We emphasize that this is not a cryptographic assumption or any form of computational hardness assumption.

Of particular interest is the fact that our security analysis makes non-black-box use of the SNARG adversary, circumventing the black-box barrier of Gentry and Wichs (STOC '11). This gives a blueprint for constructing SNARGs for NP that is not subject to the Gentry-Wichs barrier.

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1 Introduction

In the past three decades we have seen exciting progress on the problem of converting any efficiently verifiable proof into a succinct one that is only computationally sound. Specifically, the goal is to convert any witness w, corresponding to a claim $x \in L$, where L is an NP language, into a succinct certificate π certifying that $x \in L$. This succinct certificate depends on a common reference string (crs) that both the prover and the verifier have access to, and the soundness guarantee is that a computationally bounded cheating prover cannot generate a succinct certificate for a false claim. Such a system is called a *Succinct Non-interactive Argument* (SNARG), where the term argument refers to the fact that the proof is only computationally sound.

Micali [Mic94] constructed the first SNARG for NP, but its security proof was in an ideal model, called the Random Oracle Model [BR93]. Since then, there have been hundreds of papers on this topic and significant progress was made. But despite over three decades of research, several extremely basic questions about SNARGs remain unanswered. In this work, we focus on the following.

1. **Can we build SNARGs for NP based on simple, standard assumptions** such as the hardness of learning with errors (LWE)? The only constructions we have at the moment make use of extremely powerful forms of cryptography, namely, either indistinguishability obfuscation [SW14, WW24] or certain forms of witness encryption [JKLM25], both of which are quite challenging to construct [JLS21].

   One large challenge in resolving this first question is due to the fact that constructions based on LWE, and any other polynomial-time falsifiable assumption, are subject to a black-box barrier of Gentry and Wichs [GW11], which we discuss next.

2. **Is there any way around the Gentry-Wichs black-box barrier**? In a seminal work [GW11], Gentry and Wichs prove an impossibility result ruling out constructions of SNARGs based on all falsifiable hardness assumptions (including LWE, DDH, and essentially all standard cryptographic assumptions). The impossibility result is subject to the following two important caveats:

   - The result applies to constructions of *adaptively sound* SNARGs, in which the adversarial prover may choose the false statement $x^* \notin L$ adaptively as a function of the crs.
   - The result only rules out *black-box security proofs*, which are efficient black-box reductions converting any SNARG adversary into an attack on the underlying hardness assumption.

   This gives two possible approaches that might circumvent Gentry-Wichs: focus on non-adaptive soundness, or make use of a non-black-box security analysis. Unfortunately, it remains quite plausible that Gentry-Wichs extends to the non-adaptive setting [CJ25], while non-black-box techniques in cryptography are rare [Bar01, BP13, BKP18, KY18, RV22] and none have been found in the context of SNARGs.

1.1 Our Results

In this work, we give positive resolutions to the above two questions conditioned on a simple and explicit mathematical conjecture about multivariate polynomials over the real numbers. We will describe the conjecture below, but in a nutshell, it concerns *the size of the coefficients* appearing in Hilbert's Nullstellensatz for a particular asymptotic family of polynomial ideals [PZ24].

More broadly, we reopen the possibility of constructing SNARGs for NP using non-signaling probabilistically checkable proofs (nsPCPs) [KRR14], an approach that has been used to great success in constructing SNARGs for *restricted* classes of NP languages [KRR14, KP16, BHK17, BKK+18, KPY19, CJJ22, DGKV22, PP22, KLVW23, BBK+23, JKLV24] but was previously considered incapable of building full-fledged SNARGs for NP [KRR14, JKLV24].

The simplest form of our main theorem statement is as follows.

Theorem 1.1. *Suppose that the low-norm Nullstellensatz hypothesis holds for the* AND-*code (Conjecture 1.3). Then, there exist non-adaptive* SNARG*s for* NP *assuming the existence of the following two cryptographic primitives:*

- *A linearly homomorphic encryption scheme, and*
- *A somewhere extractable non-interactive batch argument system (*seBARG*) for* NP *[CJJ22, KVZ21, KLVW23].*

In particular, such a construction exists assuming either the (polynomial) hardness of LWE *[CJJ22], the k-Lin assumption on bilinear maps [WW22, KLVW23], or the quadratic residuosity and sub-exponential* DDH *assumptions [HJKS22, KLVW23, CGJ+23].*

We make the following remarks on Theorem 1.1:

- Unlike the candidate SNARG constructions in several previous works [BBK+23, JKLV24, JKLM25], our construction does not require the use of fully homomorphic encryption, which allows for additional instantiations based on number-theoretic cryptography.
- The construction and (cryptographic) security proof in Theorem 1.1 are significantly simpler compared to these prior works. In particular, our result does not require the machinery of extended Frege propositional logic, although the result was heavily influenced by this line of work.
- While our result only builds non-adaptive SNARGs, it *also* circumvents Gentry-Wichs through the use of a **non-black box security proof**. See Section 2.2 for details.
- A positive resolution to Conjecture 1.3 would, of course, give the desired SNARGs for NP. However, even if Conjecture 1.3 is false, there are several generalizations of the conjecture that we prove still suffice for a similar SNARG construction (see Theorem 5.4).

Finally, we also formulate sufficient conditions on an exponential-length PCP that would suffice for a SNARG construction under the LWE assumption.

Theorem 1.2 (informal)**.** *Suppose that there exists an exponential-length, constant-locality PCP for* NP *satisfying the following properties:*

- *Any bit of the PCP string π can be computed in polynomial time from the* NP *witness w.*
- *The PCP satisfies* weak *soundness against non-signaling strategies (Definition 7.1).*
- *The PCP has a polynomial-length extended Frege propositional proof of completeness.*

Then, there exists non-adaptive SNARGs *for* NP *under the* LWE *assumption.*

Theorem 1.2 gives a more general blueprint for constructing SNARGs for NP from LWE, because as we discuss in the technical overview, such PCPs are not subject to any impossibility result, to the best of our knowledge.

1.2 The low-norm Nullstellensatz Hypothesis

Finally, we describe our mathematical conjecture in more detail. We consider the ring of real-valued functions defined over the hypercube $\{\pm 1\}^m$, which is the polynomial ring $R = \mathbb{R}[x_1, \dots, x_m]/\langle x_1^2 - 1, \dots, x_m^2 - 1\rangle$. For this ring R, Hilbert's Nullstellensatz tells us that for any collection of polynomials $p_1(x), \dots, p_t(x)$ with common zero set $V = \{x \in \{\pm 1\}^m : p_1(x) = \cdots = p_t(x) = 0\}$, the ideal $\mathcal{I} = I(V) \subset R$ of polynomials vanishing on V is equal to $\langle p_1, \dots, p_t\rangle$, the ideal generated by the p_j. In other words, for every polynomial $f(x)$ that vanishes on V, there exist polynomials $q_1, \dots, q_t$ such that $f = \sum_j p_j \cdot q_j$.

Our conjecture concerns the following question: how *large* do the polynomials q_j have to be? Our size metric of choice is the "total coefficient size" [PZ24], i.e., the sum of the L^1 norms of the coefficient vectors

$$\sum_{j=1}^{t} ||q_j||_1.$$

For any given f, one can ask what is the *minimum* size of a Nullstellensatz decomposition $\sum_j p_j \cdot q_j$. Since f can have an arbitrary normalization, it is natural to compare this size with $||f||_1$, the L^1 norm of the original polynomial f.

We say that a collection of polynomials $p_1, \dots, p_t$ is **Nullstellensatz-friendly** if Nullstellensatz only requires a $\mathsf{poly}(m)$ blowup in coefficient size; that is, for all $f \in \mathcal{I}$, there exists a decomposition $f = \sum_j p_j \cdot q_j$ with $\sum_j ||q_j||_1 \le \mathsf{poly}(m) \cdot ||f||_1$.

Our main conjecture is that a specific ideal $\mathcal{I}$, corresponding to a specific polynomial system $\{p_j\}$, is Nullstellensatz-friendly. The setting is as follows:

- We have $m = n + \binom{n}{2}$ and variables $x_1, \dots, x_n$ along with $y_{i,j}$ for $1 \le i < j \le n$.
- For every $i < j$ we have a polynomial $p_{i,j}(x, y) = 1/2(1 - y_{i,j} \cdot (x_i \wedge x_j))$ where $x_i \wedge x_j = 1/2(1 + x_i + x_j - x_i x_j)$ computes the AND function on $\{\pm 1\}^2$.

The vanishing set of these polynomials is what we refer to as the "AND code" $\mathcal{C}_{\mathsf{AND}} = \{(x, y) \in \{\pm 1\}^{n+\binom{n}{2}} : y_{i,j} = x_i \wedge x_j \text{ for all } i < j\}$. In summary, our conjecture is as follows.

Conjecture 1.3 (Low-norm Nullstellensatz Hypothesis for the AND Code)**.** *For every polynomial $f(x, y)$ that vanishes on $\mathcal{C}_{\mathsf{AND}}$, there exist polynomials $q_{i,j}(x, y)$ such that*

$$f = \sum_{i<j} p_{i,j} \cdot q_{i,j}$$

and

$$\sum_{i<j} ||q_{i,j}||_1 \le \mathsf{poly}(n) \cdot ||f||_1$$

for some fixed polynomial function $\mathsf{poly}(n)$*.*

More generally, we state an analogous low-norm Nullstellensatz Hypothesis for *any* code $\mathcal{C}$ that is the vanishing set of a collection of ℓ-local polynomials $p_1, \ldots, p_t$. We then show that for any $\mathcal{C}$ satisfying the low-norm Nullstellensatz hypothesis and one further technical condition – the NP-completeness of k-LIN restricted to $\mathcal{C}$ – we can build a SNARG for NP as in Theorem 1.1.

We elaborate on what we know about Conjecture 1.3 in Section 5.1. Overall, we wish to highlight this conjecture to the proof complexity and broader TCS communities, identifying total coefficient size of Nullstellensatz proofs [PZ24] as having an important cryptographic application.

1.3 Organization

In Section 2 we give a technical overview of our main results, as well as a discussion of related works in Section 2.6. In Sections 3 and 4 we state the relevant preliminaries and cryptographic building blocks for our results. In Section 5 we state our low-norm Nullstellensatz hypotheses (both Conjecture 1.3 and a generalization to other codes), and formally state our main theorems. In Sections 6 and 7 we relate the low-norm Nullstellensatz hypotheses to the soundness of certain non-signaling PCPs. Finally, in Section 8 we formally state our SNARG construction (in a more generalized form) and complete the proof of our main theorems. We differ some prelimiaries to Appendix A, and the proof of Theorem 1.2 to Appendix B.

2 Technical Overview

We begin this technical overview by presenting our simplified SNARG construction.

Hash-and-BARG. The construction follows the Hash-and-BARG blueprint from prior work [CJJ22, KVZ21]. In this blueprint the prover, given a witness w corresponding to the instance x, produces a SNARG proof as follows:

1. Compute the circuit C_x on its witness w, where C_x takes as input w and outputs 1 if and only if w is a valid witness for x.[1]

2. Compute the hash of the values of all the wires in the circuit C_x, where the hash is a somewhere extractable succinct hash [HW15].

3. Compute a batch argument (BARG) that for every gate in the circuit there is a local opening of the inputs and output wire of the gate that satisfy the gate.

The succinctness of the scheme follows from the fact that the BARG grows only with the size of a single statement (and not the batch size), and each statement only consists of $O(1)$ wire values and hash opening. While the soundness of this scheme is known for some subclasses of NP, we do not know how to show it is sound for all of NP.

Variant of Encrypt-Hash-and-BARG. In our work, the circuit C_x that we consider has cryptography embedded into it and is denoted by $\widehat{C}$. It follows the recent Encrypt-Hash-and-BARG blueprint, introduced in [JKLV24, JKLM25]. However, our construction is significantly simpler and

[1] Typically, this circuit is augmented with additional gates, which is helpful in the soundness analysis.

reminiscent of the earlier work of [KRR14]. The construction is as follows. Fix any PCP scheme with locality ℓ.[2]

1. **CRS generation:** The CRS contains homomorphic encryptions of random queries, generated via the PCP verifier algorithm, where each query is encrypted via a fresh key. We denote these ciphertexts by $\widehat{q_1}, \ldots, \widehat{q_\ell}$.

2. **Prover algorithm:** Consider the circuit $\widehat{C} := \widehat{C}_{x,\widehat{q_1},\ldots,\widehat{q_\ell}}$ that on input w, does the following:

   - Compute $b = C_x(w)$.
   - For each $\widehat{q_i}$, compute $\widehat{a_i}$ by homomorphically[3] computing the PCP answer to query $\widehat{q_i}$.
   - The circuit outputs $b, \widehat{a_1}, \ldots, \widehat{a_\ell}$.

   The prover algorithm outputs $(b, \widehat{a_1}, \ldots, \widehat{a_\ell})$, along with the Hash-and-BARG proof π_{HB} corresponding to $\widehat{C}$.

3. **Verifier algorithm:** The verifier checks that $b = 1$ and that the Hash-and-BARG proof verifies.

It is easy to see that the succinctness of the proof from the locality of the PCP and the succinctness of the hash-and-BARG scheme. Therefore, this is a very natural candidate for a SNARG for NP. In this work, we ask:

Is this construction sound?

Unfortunately (but unsurprisingly), we do not know the answer to this question in general, and suspect that it is not sound for all PCPs.

Soundness. In this work, we show that this construction is indeed sound when instantiated with a variant of the Hadamard PCP (which we define in Section 2.3), assuming this Hadamard PCP satisfies a "weak non-signaling soundness" condition. Crucially, we then show that under the low-norm Nullstellensatz Hypothesis (Conjecture 1.3), which we will elaborate on in Section 2.4, the Hadamard PCP indeed has "weak non-signaling soundness."

We also prove that the SNARG construction from [JKLM25] is sound assuming the existence of *any* PCP with "weak non-signaling" soundness, which also has a polynomial length Extended Frege proof of completeness.

Organization of this overview. In Section 2.1, we begin by defining non-signaling PCPs, discussing prior work on non-signaling PCPs and SNARGs, and describing our new construction of SNARGs based on "weakly sound nsPCPs" of exponential length. In Section 2.2, we discuss the existing "barriers" to achieving SNARGs for all of NP and how our construction avoids these barriers. In Section 2.3, we define Hadamard-encoded PCPs, and in Section 2.4 we relate their (weak) non-signaling soundness to Conjecture 1.3. Finally, in Section 2.5 we discuss how to analyze soundness of our SNARG construction assuming the weak non-signaling soundness of a Hadamard-encoded PCP.

[2] This means that the verifier queries the PCP in ℓ locations.

[3] If the PCP answers are linear in q_i, then linearly homomorphic encryption is sufficient.

2.1 SNARGs from non-signaling PCPs, Revisited

Starting from the work of Kalai, Raz and Rothblum [KRR14], there has been a long line of work showing how to construct SNARGs for languages that have "non-signaling probabilistically checkable proofs" (or nsPCPs). Briefly, a nsPCP for an NP language L is a PCP for L that is sound against *non-signaling assignments*. In more detail, for a given locality parameter ℓ and a PCP of length m, an ℓ-local non-signaling assignment for the PCP is a family of distributions $\mathcal{D}_S$ over $\{0,1\}^S$ for all subsets $S \subset [m]$ of size $|S| \leq \ell$ with the property that for every pair of subsets S, T, the marginal distributions $\mathcal{D}_S|_{S \cap T}$ and $\mathcal{D}_T|_{S \cap T}$ are "close:" either identical, statistically indistinguishable, or computationally indistinguishable.[4] A nsPCP has soundness error ϵ if for all non-signaling assignments to the PCP, the probability (over a random query set) that the assignment makes the (local) verifier accept is at most ϵ.

Over the years, there have been several formal connections between nsPCPs and SNARGs. We list three important examples below.

- [KRR14] proved that, assuming the existence of fully homomorphic encryption, one can construct a *designated-verifier* SNARG for any NP language L that has a non-signaling PCP with $\mathsf{poly}\log n$ locality and negligible soundness error.

  In fact, [KRR14] show that it suffices for L to have a non-signaling multi-prover interactive proof (MIP), which we will not discuss here but turns out to be a useful relaxation.

  Roughly, in the designated verifier SNARG of [KRR14], the CRS consists of encrypted queries $\hat{q}_1, \ldots, \hat{q}_\ell$, where $q_1, \ldots, q_\ell$ are generated by the query generation algorithm of the PCP (or MIP), and the SNARG consists of the encrypted PCP (or MIP) answers $\hat{a}_1, \ldots, \hat{a}_\ell$.

- Under standard computational assumptions (such as LWE), [CJJ22, KVZ21] construct (publicly verifiable) SNARGs for any L that has a nsPCP of polynomial length and $\mathsf{poly}\log n$ locality, and a certain kind of structured query set.

  Roughly, this SNARG consists of a hash of the entire PCP and a BARG that for every possible (local) test of the PCP verifier there is an opening of the hash that makes the test accept.

- [JKLM25] show (under LWE) how to convert any designated-verifier SNARG into a publicly-verifiable SNARG, assuming that the underlying dvSNARG has an polynomial-length extended Frege propositional proof of completeness (of polynomial length).

By combining [JKLM25] and [KRR14], we obtain a new SNARG construction using any nsPCP (assuming it has polynomial-length extended Frege propositional proof of completeness).

From this technical standpoint, we aim to make progress in this work due to **two important observations**:

- We can make use of nsPCPs of *exponential* length, provided that any individual symbol of an honest PCP string can be computed in polynomial time given an NP witness.

  This holds by observing that the SNARG construction presented above only requires that the honest prover be able to homomorphically evaluate the answer to any given PCP query given its NP witness, rather than writing down the entire PCP string.

[4] As observed by [CMS18], when ℓ is a constant, all three of these notions are essentially equivalent, so we will ignore the distinction in this overview.

- We can actually make use of nsPCPs with *weaker* soundness than has been previously considered.

  Jumping ahead, this is due to the fact that the SNARG is publicly verifiable, which actually enhances security! Specifically, by the semantic security of the underlying encryption scheme, one can argue that the (cheating) prover's acceptance probability is high *for every* possible set of encrypted queries $q_1, \ldots, q_\ell$ (whereas in [KRR14] this guarantee holds only for random queries).

Weak nsPCP soundness. We now elaborate on the latter observation. We define a nsPCP to be **weakly sound** if there exists a polynomial[5] function $\mathsf{poly}(n)$ such that for any non-signaling strategy $\{\mathcal{D}_S\}_S$, *there exists a query set* S such that

$$\Pr_{\pi_S \leftarrow \mathcal{D}_S}\Big[V(S, \pi_S) = 1\Big] \leq 1 - 1/\mathsf{poly}(n).$$

In other words, weak soundness only requires that a non-signaling strategy can be caught on *a single query set*, and only with a (fixed) noticeable probability.

In general, this soundness property appears to be far weaker than standard soundness, especially in the event that the PCP has exponential length (in which case there are exponentially many possible query sets). Nevertheless, we show that the Encrypt-Hash-and-BARG construction of [JKLM25] is sound for any L possessing a nsPCP with weak soundness, provided that it has an EF proof of completeness (Appendix B). While the latter property may be somewhat opaque, it turns out to come "for free" for an important class of candidate nsPCPs that we consider for the remainder of this work.

In fact, we show that even our simplified SNARG construction described above is sound assuming the Hadamard PCP has weak non-signaling soundness, and moreover, we prove that it indeed has weak non-signaling soundness under the low-norm Nullstellensatz Hypothesis for the AND code (Conjecture 1.3).

2.2 On the non-signaling barrier and Gentry-Wichs

Before proceeding further, it is important to ask whether these relaxations buy us anything. Specifically, is there any hope for an NP-complete language to have a weakly sound, exponential-length nsPCP? To understand this question, we consider the two main barriers that have affected SNARG constructions in the past: the non-signaling barrier and the Gentry-Wichs barrier.

The **non-signaling barrier** amounts to the following negative result: if a language L has a nsPCP of length s and locality ℓ, then L can be decided in $\mathsf{DTIME}(s^\ell)$ by solving a $s^{O(\ell)}$-size linear program. Indeed, this barrier exists for nsPCPs with *any non-trivial soundness*, meaning that for all $x \notin L$, non-signaling strategies cannot pass the PCP tests with probability 1. This is because exact satisfiability of the linear program (for each x) can be decided in time $s^{O(\ell)}$.

[5] In fact, by making a 2^λ hardness assumption in the cryptographic security proof, it would even suffice to consider a weaker form of nsPCP soundness requiring that

$$\Pr_{\pi_S \leftarrow \mathcal{D}_S}\Big[V(S, \pi_S) = 1\Big] \leq 1 - 2^{-\lambda}.$$

In the past, this was considered a full-fledged no-go result for SNARGs based on nsPCPs, as it seems to contradict the exponential time hypothesis (ETH). However, we observe that this is only true for *polynomial length PCPs*! If the PCP has length $s = 2^n$ (or $s = 2^{|w|}$, where w is the witness length), this algorithm for L does not contradict ETH and does not seem to impose any barrier.

Indeed, later we **prove unconditionally** that many exponential-length PCPs have the following "ultra-weak" non-signaling soundness property: for all $x \notin L$ and all perfectly non-signaling PCP strategies, there exists a query set on which the PCP verifier rejects with positive probability. This is something ruled out by the non-signaling barrier in the case of polynomial-length PCPs, further demonstrating that exponential-length PCPs are not subject to the barrier.

Next, the **Gentry-Wichs** barrier [GW11] is an impossibility result for black-box security proofs of SNARGs for NP based on falsifiable assumptions. Since our SNARG candidates appear to have security proofs based on the (polynomial-time) hardness of LWE (assuming that nsPCPs as above exist), the Gentry-Wichs barrier may seem to suggest that the desired nsPCPs may not exist. However, there are two reasons why Gentry-Wichs does not apply to our construction and security proof.

1. Our security proof is only for *non-adaptive soundness* of the SNARG candidate, while Gentry-Wichs applies in the setting of adaptive soundness. While this means that Gentry-Wichs *technically* does not apply, we only consider this weak evidence because it remains plausible for Gentry-Wichs to extend to the non-adaptive setting [CJ25].

2. More importantly, our security proof is actually *non-black-box* in the SNARG adversary!

The fact that our security proof is non-black-box is subtle, so we elaborate on this here. The most natural formulation of our security proof is the following: "assuming the security of the underlying cryptographic primitives, any SNARG adversary can be converted into a non-signaling strategy for the underlying nsPCP. Therefore, if the nsPCP is (information-theoretically) sound, the SNARG must be computationally sound."

In order to view this proof as a reduction breaking the underlying cryptography if the SNARG is unsound, one reformulates the security proof as follows:

- There is an efficient black-box algorithm that converts a SNARG adversary into an algorithm representing an *allegedly* non-signaling strategy for the nsPCP.

- Since the nsPCP is actually sound, this implies that for any valid SNARG adversary, the (alleged) nsPCP strategy actually *fails to be non-signaling*. This means that *there exist sets* S, T on which the non-signaling property fails.

- The failure to be non-signaling on (S, T) gives rise to an attack on one of the underlying cryptographic primitives.

This security proof is actually not black-box with respect to the adversary. The reason is that the choice of sets S, T on which non-signaling fails *can depend on the adversary*, and our proof does not give any algorithm for finding a pair (S, T) on which non-signaling is violated. As a result, the security proof is non-black-box and even *non-explicit*: it proves that an attack on the underlying cryptography *exists* without actually constructing the attack.

Finally, we briefly mention that while this suggests that all SNARGs based on nsPCPs have non-black-box security proofs, the security proofs in prior work [KRR14, BHK17, BKK+18, CJJ22, JKLV24, JKLM25] are all black-box: since they have incremental "local-to-global" soundness analyses, the corresponding security reductions *do* always produce sets (S, T) on which non-signaling is violated. As a result, in order to construct a nsPCP as we want, it is necessary to make use of an "inherently global" analysis of non-signaling strategies.

2.3 Hadamard-encoded PCPs

Next, we propose a concrete family of candidate nsPCPs that we believe may have weak soundness. Later, we show how to analyze some of these PCPs under a "low-norm Nullstellensatz hypothesis."

The simplest PCP we consider is the exponential-length "Hadamard PCP" of [ALM+92]. In fact, we consider the following simplified variant of this PCP:

- We work with the following NP-complete problem:
  - NP witnesses consist of strings $w = (x, y) \in \{0,1\}^{n+\binom{n}{2}}$.
  - Instances consist of a set of 2-local linear constraints $w_i \oplus w_j = 1$.
  - A witness (x, y) is valid if it satisfies the linear constraints as well as the equations $y_{i,j} = x_i x_j$ for all pairs $i < j$.

  This is exactly the kind of constraint satisfaction problem obtained by applying a Cook-Levin reduction to an instance of Circuit-SAT.

- The PCP string is $\pi = \mathsf{Had}(w) \in \{0,1\}^{2^{n+\binom{n}{2}}}$, with $\pi_a = \langle a, w\rangle$ for each $a \in \{0,1\}^{n+\binom{n}{2}}$.
- PCP query sets consist of ℓ vectors $a_1, \ldots, a_\ell \in \{0,1\}^{n+\binom{n}{2}}$.
- The PCP verifier performs the following tests:
  - If the set contains the vectors $a, b, a \oplus b$, the verifier checks if $\pi_a \oplus \pi_b = \pi_{a\oplus b}$.
  - If the set contains standard basis vectors e_i, e_j, where (i, j) is one of the linear constraint on w (defined by the instance), the verifier checks if $\pi_{e_i} \oplus \pi_{e_j} = 1$.
  - If the set contains standard basis vectors $e_i, e_j, e_{i,j}$, the verifier checks if $\pi_{e_i} \cdot \pi_{e_j} = \pi_{e_{i,j}}$.

More generally, we can consider Hadamard encodings of an *arbitrary* PCP of polynomial length. That is, given an arbitrary polynomial-length PCP $w \mapsto \pi$ with PCP verifier V, we can consider the following exponential-length PCP:

- The new PCP string $\hat{\pi} = \mathsf{Had}(\pi)$ is the Hadamard encoding of the original PCP string π.
- PCP queries consist of vectors $a_1, \ldots, a_\ell$ as before.
- The PCP verifier performs the following checks:
  - If the query set contains $a, b, a \oplus b$, it performs the linearity test as before.
  - If the query set contains standard basis vectors $\{e_i\}_{i\in S}$ corresponding to a PCP query set S for π, the verifier performs the old PCP check on $\{\hat{\pi}_{e_i}\}_{i\in S}$.

The Hadamard PCP is exactly a Hadamard Encoding of the trivial Cook-Levin "PCP."

2.4 nsPCP soundness under low-norm Nullstellensatz

Now, for the key question: are Hadamard-encoded nsPCPs (weakly) sound? As a warm-up, we first prove that Hadamard-encoded PCPs cannot be *perfectly* satisfied by a non-signaling strategy of sufficiently large constant locality.

Lemma 2.1. *For any polynomial-length PCP with constant verifier locality ℓ, the corresponding Hadamard-encoded PCP has the following weak soundness property: for any perfectly non-signaling strategy with locality $\ell + O(1)$, there exists a query set on which the Hadamard-encoded verifier rejects with positive probability.*

The reason this lemma does not immediately solve the problem is that a priori, the verifier may only reject with probability $2^{-O(m)}$, where $m = \mathsf{poly}(n)$ is the length of the initial PCP, while we would like to prove that the verifier rejects with probability $n^{-O(1)}$.

Nevertheless, let us sketch a proof of this lemma as a starting point.

Sherali-Adams Pseudoexpectations Throughout the rest of this section (and in the body of the paper), rather than referring to ℓ-local non-signaling strategies $\{\mathcal{D}_S\}_S$, we will instead work with degree ℓ Sherali-Adams pseudoexpectations in $\{\pm 1\}$ variables $\{x_a\}_{a\in\{0,1\}^m}$. A degree ℓ pseudoexpectation $\widetilde{\mathbb{E}}$ is a linear operator mapping polynomials in $\mathbb{R}[\{w_a\}_{a\in\{0,1\}^m}]$ to the real numbers, such that:

- $\widetilde{\mathbb{E}}[1] = 1$,
- $\widetilde{\mathbb{E}}[x_a^2] = 1$ for all a ("supported on $\{\pm 1\}$"), and
- For every ℓ-local polynomial f (i.e., a polynomial that depends on only ℓ variables) that is non-negative on all inputs in $\{\pm 1\}^{2^m}$, $\widetilde{\mathbb{E}}[f] \geq 0$.

In this language, every PCP test corresponds to an ℓ-local polynomial $p((x_a)_a)$, where the test is satisfied if $p((x_a)_a) = 0$. Any ℓ-local non-signaling strategy defines a distribution on any ℓ variables, and hence a distribution on the value of any ℓ-local polynomial p. Thus, we can define $\widetilde{\mathbb{E}}[p]$ to be the expected value of p on the input distribution defined by the non-signaling distribution. Using this definition, the assumption that a non-signaling strategy *perfectly* passes a test represented by p is equivalent to the claim that $\widetilde{\mathbb{E}}[p^2] = 0$.

Proof sketch of the lemma. For a fixed false NP statement, let $p_1, \ldots, p_t$ denote polynomials that represent the constraints of the original PCP verifier. We will think of two different versions of these polynomials:

- Each p_j is a polynomial defined on variables $x_1, \ldots, x_m$ representing assignments to the original PCP.
- At the same time, we think of p_j as being a polynomial in the Hadamard-encoded variables $x_{e_1}, \ldots, x_{e_m}$.

Finally, we let $p_{a,b}(x) = \frac{1}{2}(1 - x_a x_b x_{a\oplus b})$ denote the "linearity test polynomials" defined on the Hadamard encoded variables. As stated above, our assumption is that for every PCP test p, we have $\widetilde{\mathbb{E}}[p^2] = 0$; this also implies that for any polynomial q such that $\widetilde{\mathbb{E}}[p \cdot q]$ is defined (i.e., is sufficiently low degree), we have that $\widetilde{\mathbb{E}}[p \cdot q] = 0$.

Fascinatingly, it turns out that this fact is enough to prove the lemma! The reasoning is as follows:

- Provided that the original PCP is sound, the polynomials $p_1(x), p_2(x), \ldots, p_t(x)$ have no common zero in $\{\pm 1\}^m$.
- By Hilbert's Nullstellensatz over the hypercube (formally stated in Theorem 3.23), this implies that the polynomial ideal inside of $R = \mathbb{R}[x_1, \ldots, x_m]/\langle x_1^2 - 1, \ldots, x_m^2 - 1\rangle$ generated by $p_1, \ldots, p_t$, is equal to the entire polynomial ring R.
- This implies that there exist polynomials $q_1, \ldots, q_t$ such that
$$\sum_j p_j(x) \cdot q_j(x) = 1.$$
- Finally, we argue (see Section 6.1) that this implies that there exist *constant-degree polynomials* $q_1', \ldots, q_t'$ as well as $q_{a,b}'$ such that
$$1 = \sum_j p_j(x) \cdot q_j'(x) + \sum_{a,b} p_{a,b}(x) \cdot q_{a,b}'(x).$$
- This yields a contradiction, because then we would have
$$1 = \widetilde{\mathbb{E}}[1] = \sum_j \widetilde{\mathbb{E}}[p_j(x) \cdot q_j'(x)] + \sum_{a,b} \widetilde{\mathbb{E}}[p_{a,b}(x) \cdot q_{a,b}'(x)] = 0,$$
under the assumption that all of the nsPCP tests are passed perfectly.

Weak soundness under a conjecture. Why doesn't the proof above imply weak soundness? What happens if the pseudoexpectation $\widetilde{\mathbb{E}}$ only approximately satisfies the constraints?

First of all, it turns out that (losing a constant factor in the error), we may assume without loss of generality that the linearity tests are *perfectly* satisfied by $\widetilde{\mathbb{E}}$. This is due to a beautiful analysis of Chiesa, Manohar, and Shinkar [CMS18], who prove (in our language) that any $\widetilde{\mathbb{E}}$ approximately satisfying the linearity test is "close" (in an appropriate sense) to another pseudoexpectation that perfectly satisfies the linearity tests.

Thus, we turn to the constraint polynomials $p_j(x)$. For any $\epsilon = \epsilon(n)$, $\widetilde{\mathbb{E}}$ approximately satisfying the constraints is equivalent to the statement that
$$\widetilde{\mathbb{E}}[p_j^2] \leq \epsilon.$$
What does this imply about $\widetilde{\mathbb{E}}[p_j \cdot q]$ for another low-degree polynomial q? Well, there is no fixed bound that holds for all q, because q can be scaled by an arbitrary real number. Instead, it turns out that the "right" bound is given by
$$\widetilde{\mathbb{E}}[p_j \cdot q] \leq \epsilon \cdot ||q||_1,$$

where $||q||_1$ is the sum of the absolute values of the coefficients of q. Thus, the previous analysis could be extended if we could prove that the 1 polynomial could be represented in the form $1 = \sum_j p_j(x) \cdot q_j(x)$ for polynomials $q_1, \dots, q_t$ that each have *low L^1-norm*.

Reducing to Conjecture 1.3. In general, it seems difficult to understand when such a low-norm decomposition $1 = \sum_j p_j \cdot q_j$ exists. Rather than tackling this question in general, we now focus on the special case of the Hadamard PCP. In the body, we extend this special analysis to a broader class of Hadamard-encoded PCPs.

In the case of the Hadamard PCP, we are considering polynomials in the variables $x_1, \dots, x_n$ and variables $y_{i,j}$ for $i < j$. The constraint polynomials are as follows:

- $p_{i,j}(x, y) = 1/2(1 - y_{i,j} \cdot (x_i \wedge x_j))$, where $\wedge$ denotes the $\{\pm 1\}$-valued Boolean AND function
- Linear (mod 2) constraints $1/2(1 + y_{i,j} \cdot x_k)$ for triples (i, j, k) defining the NP instance.

We will now argue soundness of this nsPCP under Conjecture 1.3, a conjecture about the *single* polynomial ideal $\mathcal{I} = \langle p_{i,j} \rangle$, independent of the choice of false NP instance. Recall that the conjecture is that polynomials in $\mathcal{I}$ admit low-norm Nullstellensatz decompositions; that is, for every $f \in \mathcal{I}$,

$$f = \sum_{i,j} p_{i,j} \cdot q_{i,j}$$

for some polynomials $\{q_{i,j}\}$ such that

$$\sum_{i,j} ||q_{i,j}||_1 \leq \mathsf{poly}(n) \cdot ||f||_1.$$

Under this single conjecture, we are able to prove soundness of the nsPCP on *every* false instance. To do this, we identify special polynomials

$$p^*(x, y) = \frac{1}{2^t} \sum_{T \subset [t]} \frac{1 - (-1)^{|T|} \cdot \prod_{\ell \in T} y_{i_\ell, j_\ell} \cdot x_{k_\ell}}{2}$$

defined with respect to a collection of unsatisfiable linear constraints, such that

- Each $p^*(x, y)$ vanishes on the points $(x, y) \in \{\pm 1\}^m$ satisfying the $p_{i,j}$. By Nullstellensatz, this implies that $p^*(x, y) \in \mathcal{I}$.
- The norm $||f||_1 = O(1)$ is small. Under the conjecture, this implies that f admits a low-norm Nullstellensatz decomposition. This can be shown to imply that $\widetilde{\mathbb{E}}[f] \leq \epsilon \cdot \mathsf{poly}(n)$.[6]
- At the same time, we prove that $\widetilde{\mathbb{E}}[f] = \Omega(1)$ because of how f is built out of the linear constraints that we assume are satisfied by $\widetilde{\mathbb{E}}$.

In total, this yields a contradiction for a sufficiently small $\epsilon = n^{-O(1)}$, yielding the desired soundness property.

[6]Since f is a high degree polynomial, $\widetilde{\mathbb{E}}[f]$ is not defined a priori. However, every f can be "linearized" in the Hadamard-extended variables, so by $\widetilde{\mathbb{E}}[f]$ we mean the pseudoexpectation of the linearization of f.

2.5 Analysis of our SNARG

Suppose there exists a Hadamard-encoded PCP which is weakly non-signaling sound, where the underlying PCP has length $N = \mathsf{poly}(n)$ and locality $\ell = O(1)$. Here n is the length of an instance. We now show that our simplified Encrypt-Hash-and-BARG construction (described in the beginning of Section 2) is sound when instantiated with such the Hadamard encoded PCP. Note that since the Hadamard PCP is a linear PCP, in the SNARG construction we only need a linearly homomorphic encryption, denoted by LHE.

First, we will make a small modification to specialize the construction for the Hadamard PCP so that we will only need *linearly homomorphic encryption* (LHE) (rather than fully homomorphic encryption).

- **Prover algorithm:** Consider the circuit $\widehat{C} := \widehat{C}_{x,\widehat{q_1},\ldots,\widehat{q_\ell}}$ that takes as input w and does the following:
  - Compute the PCP string $\Gamma \leftarrow \mathsf{PCP}.\mathcal{P}(x, w)$.
  - Let $\mathcal{Q} = \{Q_i\}$ (note that there are at most N^ℓ such sets) be the set of all possible PCP queries. For each Q_i, check if $\mathsf{PCP}.\mathcal{V}(Q_i, \Gamma|_{Q_i}) = 1$. Set $b = 1$ if all the tests pass.
  - For each $\widehat{q_i}$, homomorphically compute Hadamard answers $\boxed{\langle\Gamma, q_i\rangle}$. Let $\widehat{a_i}$ correspond to the evaluated answer.
  - Output $b, \widehat{a_1}, \ldots, \widehat{a_\ell}$.

  The prover outputs $b, \widehat{a_1}, \ldots, \widehat{a_\ell}$, along with the Hash-and-BARG proof π_{HB} corresponding to $\widehat{C}$.

Consider a cheating prover $\mathcal{P}^*$ who produces accepting proofs for $x^* \notin \mathcal{L}$. For the sake of simplicity, in this overview we assume that $\mathcal{P}^*$ succeeds with probability $1 - \mathsf{negl}(\lambda)$.

Generating Non-Signaling Adversary for nsPCP. We now give an informal overview on how to construct a NS strategy for the Hadamard PCP that violates weak NS soundness. For a given a set of Hadamard queries $\mathcal{Q} = \{q_1, \ldots, q_\ell\}$, the distribution $\mathcal{D}_\mathcal{Q}$ works as follows:

- Generate a crs where the LHE encryptions correspond to encryptions of $\widehat{q_i} \leftarrow \mathsf{LHE.Enc}(\mathsf{sk}_i, q_i)$.
- Obtain $(b, \widehat{a_1}, \ldots, \widehat{a_\ell}, \pi_{\mathsf{HB}})$ from the prover.
- Let $a_i \leftarrow \mathsf{LHE.Dec}(\mathsf{sk}_i, \widehat{a_i})$.
- Output $\{a_i\}_{i\in[\ell]}$.

The fact that the family of distributions $\{\mathcal{D}_\mathcal{Q}\}_\mathcal{Q}$ satisfies (computationally) non-signaling follows from the LHE semantic security (as was argued in [KRR14]). It therefore remains to show that $\mathcal{D}_\mathcal{Q}$ passes the Hadamard PCP's verifier checks with probability $1 - \mathsf{negl}(\lambda)$, i.e., the following hold with probability $1 - \mathsf{negl}(\lambda)$:

- **Linear consistency:** If any subset $\{q_i, q_j, q_k\}$ of $\mathcal{Q}$ satisfies $q_i \oplus q_j \oplus q_k = \mathbf{0}$, the corresponding answers satisfy $a_i \oplus a_j \oplus a_k = 0$.
- **PCP test consistency:** If $\mathcal{Q}$ contains queries of the form $q_1 = e_{i_1}, \ldots q_\ell = e_{i_\ell}$ where $i_1, \ldots, i_\ell \in [N]$ correspond to indices of Γ, then $\mathsf{PCP}.\mathcal{V}(x, i_1, \ldots, i_\ell, a_1, \ldots, a_\ell) = 1$.

Local assignment generator via Hash-and-BARG. To show that the linear and constraint checks pass, we will rely on the "local soundness" of Hash-and-BARG. As shown in many prior works [CJJ22, KVZ21, BBK+23], if a prover $\mathcal{P}^*$ is able to create accepting Hash-and-BARG proofs[7] for a circuit C, then one can use $\mathcal{P}^*$ to construct a *local assignment generator* [PR17] which produces *consistent* partial assignments to the wires of the circuit C. Moreover, if the algorithm is queried on sets T_0 or T_1, the marginal distribution of the wire assignments on $T_0 \cap T_1$ is non-signaling.

Concretely, we show that one can use $\mathcal{P}^*$ to construct a local assignment generator $\mathsf{LocalGen}$ that takes as input a set a queries $\mathcal{Q} = \{q_1, \dots, q_\ell\}$ and subset of wires T of size at most $K = \mathsf{poly}(\lambda, 2^\ell)$, and outputs (with probability $1 - \mathsf{negl}(\lambda)$):

- a list of encrypted answers $\overrightarrow{\hat{a}} = \{\widehat{a_1}, \dots, \widehat{a_\ell}\}$,
- a list of secret keys $\overrightarrow{\mathsf{sk}} = \{\mathsf{sk}_1, \dots, \mathsf{sk}_\ell\}$,
- a wire assignment $\{\sigma_w\}_{w \in T}$ to wires T in $\widehat{C}_{x^*, \widehat{q_1}, \dots, \widehat{q_\ell}}$. Moreover, if T contains the output wire, the output value is 1.

We show that this $\mathsf{LocalGen}$ satisfies the following properties:

- **Local consistency:** For every set of ℓ wires T, the wire assignment $\{\sigma_w\}_{w \in T}$, output by $\mathsf{LocalGen}(Q, T)$, is consistent with respect to $\widehat{C}_{x^*, \widehat{q_1}, \dots, \widehat{q_\ell}}$, and is also consistent with $\overrightarrow{\hat{a}}$.
- **Indistinguishable marginals on $\overrightarrow{\hat{a}}, \overrightarrow{\mathsf{sk}}$:** Consider sets T_0, T_1 of size at most ℓ. The marginal distribution on $\overrightarrow{\hat{a}}, \overrightarrow{\mathsf{sk}}$ when sampled from $\mathsf{LocalGen}(Q, T_0)$ and $\mathsf{LocalGen}(Q, T_1)$ is computationally indistinguishable. Additionally, we set things up so that the distribution which samples $\overrightarrow{\hat{a}}, \overrightarrow{\mathsf{sk}} \leftarrow \mathsf{LocalGen}(\mathcal{Q}, \varnothing)$ and outputs $\{\mathsf{LHE.Dec}(\mathsf{sk}_i, \widehat{a_i})\}_{i \in [\ell]}$ is indistinguishable from the distribution output by $\mathcal{D}_{\mathcal{Q}}$ defined above (this corresponds to the "correct distribution" property of quasi-arguments Definition 4.10).
- **Non-Signaling on wires:** Let T_0 and T_1 be subsets of wires of C of size at most ℓ. Then, the marginal distributions of $\mathsf{LocalGen}(Q, T_0)$ and $\mathsf{LocalGen}(Q, T_1)$ on $T_0 \cap T_1$ are computationally indistinguishable.

Before showing why the above properties imply linear and constraint consistency, it is helpful to pause and consider exactly how for each query q the homomorphic evaluation of $\langle \Gamma, q \rangle$ is computed (recall that Γ is the PCP computed by the prover algorithm for instance x and witness w.) The query q is encrypted bit-by-bit (i.e. $\widehat{q} = \{\widehat{q_j}\}_{j \in [N]}$), and the inner product is computed iteratively as follows:

1. Initialize $S = 0$.
2. For each $j \in [N]$, compute the partial sum $S \leftarrow \mathsf{LHE.Add}(S, \Gamma_j \cdot \widehat{q_j})$.

With this explicit evaluation circuit in hand, we now show how to use $\mathsf{LocalGen}$ to argue that the $\overrightarrow{\hat{a}}$ passes both the linear and constraint tests.

[7] This is in fact a quasi-argument scheme as defined by [KPY19]. We work with this abstraction in our main body, and formalize the security guarantees of this abstraction in Definition 4.10, and show that hash-and-BARG is a quasi-argument in Appendix A.

Linear consistency. Suppose $\mathcal{Q}$ contains queries $\alpha, \beta, \gamma \in \{0,1\}^N$ such that $\gamma = \alpha \oplus \beta$. We use the two properties of $\mathsf{LocalGen}$ (i.e., non-signaling and local consistency) to prove by induction on $j \in [N]$ that if we extract the partial sum at step $j \in [N]$, denoted by $S_{\alpha,j}, S_{\beta,j}, S_{\gamma,j}$, then the wire assignment output by $\mathsf{LocalGen}$ satisfies that

$$\mathsf{LHE.Dec}(\mathsf{sk}_\alpha, S_{\alpha,j}) \oplus \mathsf{LHE.Dec}(\mathsf{sk}_\beta, S_{\beta,j}) \oplus \mathsf{LHE.Dec}(\mathsf{sk}_\gamma, S_{\gamma,j}) = 0. \tag{2.1}$$

The linear consistency follows by Equation (2.1) with $j = N$. We give an informal sketch of the inductive argument. For $j = 1$ this constraint is local, and hence follows from the local consistency condition. Suppose Equation (2.1) holds for $j - 1$. Consider querying $\mathsf{LocalGen}$ on the values

$$S_{\alpha,j-1}, S_{\beta,j-1}, S_{\gamma,j-1}, S_{\alpha,j}, S_{\beta,j}, S_{\gamma,j}.$$

By non-signaling, and by our induction hypothesis, $S_{\alpha,j-1}, S_{\beta,j-1}, S_{\gamma,j-1}$ satisfies Equation (2.1). The local consistency property guarantees that

$$S_{\alpha,j} = \mathsf{LHE.Add}(S_{\alpha,j-1}, \Gamma_j \cdot \widehat{\alpha_j})$$

and the same holds for β, γ as well. Therefore, by correctness of LHE, we conclude that Equation (2.1) also holds for j.

PCP test consistency. Suppose $\mathcal{Q}$ contains $q_1 = e_{i_1}, \ldots, q_\ell = e_{i_\ell}$, where $(i_1, \ldots, i_\ell)$ is in the support of $\mathsf{PCP}.\mathcal{Q}$. Let T contain the wires computing $\{\Gamma_{i_1}, \ldots, \Gamma_{i_\ell}\}$. We claim that $(\overrightarrow{\hat{a}}, \overrightarrow{\mathsf{sk}}, \{\Gamma_{i_1}, \ldots, \Gamma_{i_\ell}\}) \leftarrow \mathsf{LocalGen}(\mathcal{Q}, T)$ satisfies that $a_j = \Gamma_{i_j}$ for every $j \in [\ell]$. By the non-signaling condition we know that the PCP verifier accepts $(\Gamma_{i_1}, \ldots, \Gamma_{i_\ell})$ w.r.t. the queries $(i_1, \ldots, i_\ell)$. Thus, it indeed suffices to prove that $a_j = \Gamma_{i_j}$. The proof strategy follows inductively just as in the case of linearity tests.

2.6 Related Work

Non-signaling PCPs. As discussed earlier, our SNARG is based on a line of research relating cryptographic proof systems to non-signaling PCPs (including [KRR14, CJJ22, KVZ21, BBK+23, JKLV24, JKLM25]). However, our nsPCP analysis is a significant departure from previous nsPCP analyses in the literature [KRR14, BHK17, BKK+18] that were used for SNARG constructions. Instead, our analysis is inspired by a fascinating line of work of Chiesa, Manohar, and Shinkar [CMS18, CMS19, CMS20] as well as Jahanara, Koroth, and Shinkar [JKS20] attempting to understand constant-locality nsPCPs for P (rather than NP). We briefly compare our work with this line of work:

- On one hand, [CMS18, CMS19, CMS20, JKS20] wish to construct nsPCPs with *constant* soundness error, matching what is typically studied in the PCP literature as well as for previous cryptographic applications. This is a more challenging setting than ours.
- On the other hand, we wish to prove soundness for all of NP, while this line of work focused on the sub-class P (where, e.g., [KRR14] have an alternative, polynomial-length nsPCP).

Moreover, one can view our main conjecture as asking about the non-signaling soundness of a simple local test of the *non-linear*, exponential-length code consisting of the evaluation of all quadratic functions on a given message. [CMS20] studied local testability of *linear* codes, but their techniques are limited to this linear setting.

Nullstellensatz Proofs. Nullstellensatz has been used as a tool and object of study in proof complexity dating back to the seminal work of Beame et al. [BIK+94]. However, most of the literature on Nullstellensatz proofs focused on understanding the *degree* of the polynomials q_j required in a Nullstellensatz decomposition $f = \sum_j p_j \cdot q_j$ [BCE+95], or the *number of monomials* required in the proof [IPS99].

On the other hand, we are interested in the *total coefficient size* of Nullstellensatz certificates, exactly as formulated by a recent work of Potechin and Zhang [PZ24]. They prove an exponential coefficient size lower bound for a specific family of polynomials over $\{0, 1\}$-inputs (corresponding to the Pigeonhole Principle) that span the entire polynomial ring (as the polynomial system is unsatisfiable). As far as we can tell, this lower bound does not seem to relate to the polynomials in our conjectures. See Section 5 for more details on our conjecture.

3 Preliminaries

Notation 3.1. *For $n \in \mathbb{N}$, we write $[n]$ to denote $\{1, \ldots, n\}$. We write $\binom{[n]}{2}$ to denote the set of subsets of $[n]$ of size 2, i.e., $\{S : S \subseteq [n], |S| = 2\}$.*

Definition 3.2 (Coefficient L^1 norm)**.** Let $f(x_1, \ldots, x_n) \in \mathbb{R}[x_1, \ldots, x_n]$ be a multivariate polynomial. We define the coefficient L^1-norm of f, or $||f||_1$, to be the sum of the absolute values of the coefficients of f.

3.1 Non-signaling distribution families

Notation 3.3. *For any two distributions X and Y we use $X \overset{\delta}{\equiv} Y$ to denote the fact that the two distributions are δ-close in total variation distance.*

Fact 3.4. *Given two distributions $\mathcal{D}_0$ and $\mathcal{D}_1$ on $\{0, 1\}^k$, with total variation distance δ, there exists a $\mathsf{poly}(2^k)$-size distinguisher $\mathcal{A}$ such that:*

$$\left| \Pr_{x \leftarrow \mathcal{D}_0}[\mathcal{A}(x) = 1] - \Pr_{x \leftarrow \mathcal{D}_1}[\mathcal{A}(x) = 1] \right| = \delta.$$

Definition 3.5. We call a family of local distributions $\{D_I\}_{I \subseteq [N], |I| \le k}$ (k, δ)-non-signaling if for every $I, J \subseteq [N]$ of size $\le k$,

$$D_I|_{I \cap J} \overset{\delta}{\equiv} D_J|_{I \cap J}.$$

Theorem 3.6 ([CMS18, Lemma C.3])**.** *For every (k, δ)-non-signaling family of local distribution $\{D_I\}_{I \subseteq [N], |I| \le k}$, there exists a $(k, 0)$-non-signaling family of local distributions $\{D'_I\}_{I \subseteq [N], |I| \le k}$ such that for all $I \subseteq [N]$ where $|I| \le k$,*

$$D_I \overset{\eta}{\equiv} D'_I$$

for $\eta \le 2^k \delta + 2^{2k} \delta$.

3.2 Sherali-Adams Pseudoexpectations

Notation 3.7. *In what follows we denote by $\mathbb{R}[x_1, \ldots, x_N]_{\le k}$ the set of all multivariate polynomials of total degree at most k over the reals in variables $(x_1, \ldots, x_N)$.*

In this work, we focus on polynomial functions restricted to the domain of $\{\pm 1\}^N$, which we denote by

$$\mathbb{R}[x_1, \ldots, x_N]_{\leq k}|_{\{x_i^2 = x_i\}_{i \in [N]}}$$

and denote it by $\mathbb{R}[x]_{\leq k}|_{x^2=x}$ for short, where $x = (x_1, \ldots, x_N)$.

Definition 3.8 (k-junta)**.** A polynomial $f \in \mathbb{R}[x]_{\leq k}$ is defined to be a k-junta if $f(x)$ depends on at most k coordinates of x.

Notation 3.9. *Let f be a k-junta that depends only on a variable subset $I \subset [n]$. For any vector $a \in \{\pm 1\}^N$, we let $a_I \in \{\pm 1\}^I$ denote the restriction of a to I, and we define $f(a_I) = f(a)$. Since f is an I-junta, $f(a_I)$ is a well-defined function of a_I.*

Definition 3.10 (Monomial)**.** For every subset $I \subset [N]$, we define the monomial $m_I(x) = \prod_{i \in I} x_i$. If $|I| \leq k$, this is a k-junta representing a function $m_I : \{\pm 1\}^N \to \{\pm 1\}$.

Definition 3.11 (Non-negative k-junta)**.** A polynomial $f \in \mathbb{R}[x]_{\leq k}$ is defined to be a *non-negative* k-junta if $f(x)$ depends on at most k coordinates of x, and $f(a) \geq 0$ for all $a \in \{\pm 1\}^N$.

Definition 3.12 (Indicator k-junta)**.** For every set $I \subset [N]$ of size k and every vector $b \in \{\pm 1\}^I$, we define the indicator k-junta $f_{I,b}$ as follows:

$$f_{I,b}(x) = \prod_{i \in I} \left(\frac{1 + b_i x_i}{2} \right).$$

We observe that for every (I, b), $f_{I,b}$ is a non-negative k-junta that evaluates to 1 on all inputs a such that $a_i = b_i$ for all $i \in I$, and evaluates to 0 on all other inputs in $\{\pm 1\}^N$.

Fact 3.13. *Every non-negative k-junta is a non-negative real linear combination of indicator k-juntas with respect to a fixed set I of size k.*

Definition 3.14. A degree-k Sherali-Adams (k-SA) pseudoexpectation $\widetilde{\mathbb{E}}$ over variables $x = (x_1, \ldots, x_N)$ is an operator from $\mathbb{R}[x]_{\leq k}$ to $\mathbb{R}$ that satisfies the following:

1. **Linearity:** $\widetilde{\mathbb{E}}[ap + bq] = a\widetilde{\mathbb{E}}[p] + b\widetilde{\mathbb{E}}[q]$ for any polynomials $p, q \in \mathbb{R}[x]_{\leq k}$ and any scalars $a, b \in \mathbb{R}$.
2. **Normalization:** $\widetilde{\mathbb{E}}[1] = 1$.
3. **Non-negativity:** For every non-negative k-junta $f \in \mathbb{R}[x]_{\leq k}|_{x^2=x}$, we have $\widetilde{\mathbb{E}}[f] \geq 0$.

By Fact 3.13, the non-negativity condition above holds if and only if it holds with respect to all indicator k-juntas.

Definition 3.15. Let $\widetilde{\mathbb{E}}$ be a k-SA pseudo-expectation over variables $x = (x_1, \ldots, x_N)$, and let $p \in \mathbb{R}[x]_{\leq k}$ be an ℓ-junta for some $\ell \leq k$. We say that $\widetilde{\mathbb{E}} \vDash p = 0$ ("$\widetilde{\mathbb{E}}$ satisfies $p = 0$") if, for every $q \in \mathbb{R}[x]$ such that $\deg(q) \leq k - \ell$,[8] we have $\widetilde{\mathbb{E}}[pq] = 0$.

More generally, for a family $\mathcal{P}$ of polynomials in $\mathbb{R}[x]_{\leq k}$, we say that $\widetilde{\mathbb{E}} \vDash \mathcal{P} = 0$ if $\widetilde{\mathbb{E}} \vDash p = 0$ for every $p \in \mathcal{P}$.

[8] When q is a monomial, this corresponds to the condition that $q \cdot p$ has locality at most k.

Remark 3.16. *In this work, we only consider pseudoexpectations $\widetilde{\mathbb{E}}$ such that $\widetilde{\mathbb{E}} \vDash \mathcal{P} = 0$ for $\mathcal{P} = \{p_i\}_{i\in[N]}$ where $p_i = x_i^2 - 1$. Such $\widetilde{\mathbb{E}}$ is said to be supported on $\{\pm 1\}$.*

Definition 3.17. A family of local distributions $\{D_I\}_{I\subseteq[N],|I|\leq k}$ is said to be k-local non-signaling over $[N]$ if for every $I, J \subseteq [N]$ of size $\leq k$,

$$D_I|_{I\cap J} \equiv D_J|_{I\cap J}. \tag{3.1}$$

It is well known that k-SA pseudoexpectations over variables $(x_1, \ldots, x_N)$ supported on $\{\pm 1\}$ are equivalent to k-local non-signaling distributions over $\{\pm 1\}$. This is formalized in the following claim.

Claim 3.18 ([CMS18], Section 8)**.** *For every k-SA pseudoexpectation $\widetilde{\mathbb{E}}$ over variables $(x_1, \ldots, x_N)$ that is supported on $\{\pm 1\}$ there exists a family of k-local non-signaling distributions $\{D_I\}_{I\subseteq[N],|I|\leq k}$ such that for every junta $f \in \mathbb{R}[x_1, \ldots, x_N]_{\leq k}$ that depends only on $x_I = \{x_i\}_{i\in I}$ for $|I| \leq k$,*

$$\widetilde{\mathbb{E}}[f] = \sum_{b_I\in\{\pm 1\}^{|I|}} \Pr_{x_I\leftarrow D_I}[x_I = b_I] \cdot f(b_I) \tag{3.2}$$

Conversely, for every family of k-local non-signaling distributions $\{D_I\}_{I\subseteq[N],|I|\leq k}$ there exists a pseudoexpectation $\widetilde{\mathbb{E}}$ that satisfies Equation (3.2).

For completeness, we prove Claim 3.18 below

Proof of Claim 3.18. Fix any k-pseudoexpectation $\widetilde{\mathbb{E}}$ supported on $\{\pm 1\}$. For any $I \subseteq [N]$ of size $\leq k$, define D_I as follows: For every $b_I \in \{\pm 1\}^{|I|}$, let

$$\Pr_{x_I\leftarrow D_I}[x_I = b_I] = \widetilde{\mathbb{E}}[f_{I,b_I}], \tag{3.3}$$

where f_{I,b_I} is the indicator junta associated with the assignment b_I. The fact that f_{I,b_I} is a non-negative junta over $\{\pm 1\}$ implies these probabilities are indeed non-negative. Moreover, for every $I \subseteq [N]$,

$$\sum_{b_I\in\{0,1\}^{|I|}} \Pr_{x_I\leftarrow D_I}[x_I = b_I] = \sum_{b_I\in\{0,1\}^{|I|}} \widetilde{\mathbb{E}}[f_{I,b_I}] = \widetilde{\mathbb{E}}\left[\sum_{b_I\in\{0,1\}^{|I|}} f_{I,b_I}\right] = \widetilde{\mathbb{E}}\,[1] = 1,$$

where the equations follow from the definition of a pseudoexpectation and the fact that $\sum_{b_I} f_{I,b_I} \equiv 1$.

Next we prove that for any $I, J \subseteq [N]$ of size $\leq k$, and any $b_{I\cap J} \in \{\pm 1\}^{|I\cap J|}$,

$$\Pr_{x_I\leftarrow D_I}[(x_I)|_{I\cap J} = b_{I\cap J}] = \Pr_{x_J\leftarrow D_J}[(x_J)|_{I\cap J} = b_{I\cap J}]$$

To this end, we note that

$$\begin{aligned}
\Pr_{x_I \leftarrow D_I}[(x_I)|_{I\cap J} = b_{I\cap J}] &= \sum_{b'_I\colon\, b'_{I\cap J} = b_{I\cap J}} \Pr_{x_I \leftarrow D_I}[x_I = b'_I] \\
&= \sum_{b'_I\colon\, b'_{I\cap J} = b_{I\cap J}} \widetilde{\mathbb{E}}[f_{I,b'_I}] \\
&= \widetilde{\mathbb{E}}\left[\sum_{b'_I\colon\, b'_{I\cap J} = b_{I\cap J}} f_{I,b'_I}\right] \\
&= \widetilde{\mathbb{E}}\left[f_{I\cap J, b_{I\cap J}}\right]
\end{aligned}$$

By symmetry, this implies that $\Pr_{x_I \leftarrow D_I}[(x_I)|_{I\cap J} = b_{I\cap J}] = \Pr_{x_J \leftarrow D_J}[(x_J)|_{I\cap J} = b_{I\cap J}]$.

Finally, fix any junta $f \in \mathbb{R}[x_1, \ldots, x_N]_{\le k}$ that depends only on $x_I = \{x_i\}_{i\in I}$ for $|I| \le k$. Then

$$f(x_1, \ldots, x_N) = \sum_{b_I \in \{\pm 1\}^{|I|}} f_{x_I, b_I}(x_1, \ldots, x_N) \cdot f(b_I),$$

which by linearity implies that

$$\widetilde{\mathbb{E}}[f] = \sum_{b_I \in \{\pm 1\}^{|I|}} \widetilde{\mathbb{E}}[f_{x_I, b_I}] \cdot f(b_I) = \Pr_{x_I \leftarrow D_I}[x_I = b_I] \cdot f(b_I),$$

as desired.

For the converse, fix any family of k-local non-signaling distributions $\{D_I\}_{I\subseteq[N], |I|\le k}$. We define an associated pseudoexpectation $\widetilde{\mathbb{E}}$ to be the unique pseudoexpectation operator satisfying

$$\widetilde{\mathbb{E}}[m_I(x)] = 1 - 2 \cdot \Pr_{x_I \leftarrow D_I}\left[\prod_{i\in I} x_i = 1\right] = \mathop{\mathbb{E}}_{x_I \leftarrow D_I}\Big[m_I(x_I)\Big].$$

for all $I \subset [N]$ of size at most k. By convention, we define $\widetilde{\mathbb{E}}[1] = 1$. This uniquely defines a linear operator from $\mathbb{R}[x]_{\le k} \to \mathbb{R}$ by linearity.

To show the non-negativity property, we observe that it suffices to show non-negativity on indicator k-juntas f_{I,b_I}. To do this, we make use of the Fourier expansion

$$f_{I,b_I}(x) = \frac{1}{2^{|I|}} \sum_{J\subset I} \prod_{i\in J} b_i \cdot m_J(x).$$

By the non-signaling property, we see that

$$\begin{aligned}
\widetilde{\mathbb{E}}[f_{I,b_I}] &= \frac{1}{2^{|I|}} \cdot \sum_{J\subset I} \prod_{i\in J} b_i \cdot \mathop{\mathbb{E}}_{x_J \leftarrow D_J}\Big[m_J(x_J)\Big] \\
&= \frac{1}{2^{|I|}} \cdot \sum_{J\subset I} \prod_{i\in J} b_i \cdot \mathop{\mathbb{E}}_{x_I \leftarrow D_I}\Big[m_J(x_I)\Big] \\
&= \mathop{\mathbb{E}}_{x_I \leftarrow D_I}\Big[f_{I,b_I}(x)\Big],
\end{aligned}$$

which is indeed non-negative. $\square$

Using Claim 3.18, we observe the following fact.

Claim 3.19. *Let $\widetilde{\mathbb{E}}$ be a k-SA pseudoexpectation over variables $x = (x_1, \ldots, x_N)$, and let $p \in \mathbb{R}[x]_{\leq k}$ be a d-junta for $d \leq k$. If $\widetilde{\mathbb{E}}\left[p^2\right] = 0$ then $\widetilde{\mathbb{E}} \vDash p = 0$.*

Proof. We need to prove that for every $q \in \mathbb{R}[x]_{\leq k}$ such that $\deg(q) \leq k - \ell$ it holds that $\widetilde{\mathbb{E}}[p \cdot q] = 0$. By linearity, it suffices to prove that for every $d \leq k - \ell$ and every monomial $x_{i_1} \cdot \ldots \cdot x_{i_d}$ it holds that $\widetilde{\mathbb{E}}[p \cdot x_{i_1} \cdot \ldots \cdot x_{i_d}] = 0$. Suppose that p depends on the variables $\{x_{j_1} \ldots, x_{j_\ell}\}$ and denote by $J = \{j_1, \ldots, j_\ell\} \subseteq [N]$. By Claim 3.18, $\widetilde{\mathbb{E}}$ is associated with a set of local distributions $\{D_I\}_{I \subseteq [N], |I| \leq k}$, such that

$$\widetilde{\mathbb{E}}[p^2] = \sum_{b_J} \Pr_{x_J \leftarrow D_J}[x_J = b_J] \cdot p^2(b_J),$$

which together with our assumption that $\widetilde{\mathbb{E}}[p^2] = 0$, implies that

$$\Pr_{b_J \leftarrow D_J}[p(b_J) = 0] = 1.$$

Let $I = \{i_1, \ldots, i_\ell\} \cup J$, and note that $|I| \leq k$. Thus,

$$\Pr_{b_I \leftarrow D_I}[p((b_I)|_J) \cdot b_{i_1} \ldots \cdot b_{i_\ell}) = 0] = 1$$

which implies that $\widetilde{\mathbb{E}}[p \cdot x_{i_1} \cdot \ldots \cdot x_{i_\ell}] = 0$. $\square$

Finally, we also use Claim 3.18 to define what it means for two pseudoexpectations to be *δ-close*.

Definition 3.20 (Closeness of pseudoexpectations)**.** Two k-SA pseudoexpectations $\widetilde{\mathbb{E}}$ and $\widetilde{\mathbb{E}}'$ over the same set of variables are said to be δ-close, denoted by $\widetilde{\mathbb{E}} \overset{\delta}{\equiv} \widetilde{\mathbb{E}}'$, if the set of local distributions $\{D_I\}$ and $\{D'_I\}$ corresponding to $\widetilde{\mathbb{E}}$ and $\widetilde{\mathbb{E}}'$ have the property that for every I, $D_I \overset{\delta}{\equiv} D'_I$.

Equivalently, $\widetilde{\mathbb{E}} \overset{\delta}{\equiv} \widetilde{\mathbb{E}}'$ if for every $\{0,1\}$-valued k-junta f, $\left|\widetilde{\mathbb{E}}[f] - \widetilde{\mathbb{E}}'[f]\right| \leq \delta$.

Fact 3.21. $\widetilde{\mathbb{E}} \overset{\delta}{\equiv} \widetilde{\mathbb{E}}'$ *if either of the following two conditions is true:*

1. *For every size-k set $I \subset [N]$ and every $b \in \{\pm 1\}^I$,*

$$\left|\widetilde{\mathbb{E}}[f_{I,b_I}] - \widetilde{\mathbb{E}}'[f_{I,b_I}]\right| \leq 2^{-k} \cdot \delta$$

2. *For every size k subset $I \subset [N]$,*

$$\left|\widetilde{\mathbb{E}}[m_I] - \widetilde{\mathbb{E}}'[m_I]\right| \leq 2 \cdot 2^{-k/2} \cdot \delta$$

Fact 3.22. *If $\widetilde{\mathbb{E}} \overset{\delta}{\equiv} \widetilde{\mathbb{E}}'$, then for every degree k polynomial $f(x_1, \ldots, x_n)$, we have that*

$$\left|\widetilde{\mathbb{E}}[f] - \widetilde{\mathbb{E}}'[f]\right| \leq 2\delta \cdot ||f||_1.$$

3.3 Nullstellensatz over the Hypercube

Theorem 3.23 (Nullstellensatz over the Hypercube)**.** *Let $p_1, \ldots, p_t \in \mathbb{R}[x_1, \ldots, x_N]/\langle x_i^2 - 1\rangle_i$ be a collection of t multilinear polynomials in N variables, and let $\mathcal{P} = \{p_1, \ldots, p_t\}$.*

Let $V = V(\mathcal{P}) \subset \{\pm 1\}^N$ denote the common zero locus of $p_1, \ldots, p_t$. Then, a polynomial f vanishes on V if and only if $f \in \langle \mathcal{P} \rangle$, the ideal generated by $\mathcal{P}$.

4 Cryptographic Building Blocks

Throughout, we will use λ to denote the security parameter. We say that a function $f(\lambda)$ is negligible in λ if $f(\lambda) = \lambda^{-\omega(1)}$, and we denote it by $f(\lambda) = \mathsf{negl}(\lambda)$.We say that a function $g(\lambda)$ is polynomial in λ if $g(\lambda) = p(\lambda)$ for some fixed polynomial p, and we denote it by $g(\lambda) = \mathsf{poly}(\lambda)$.

Notation 4.1. *If R is a random variable, then $r \leftarrow R$ denotes sampling r from R. If T is a set, then $i \leftarrow T$ denotes sampling i uniformly at random from T. If $\mathcal{A}$ is a (possibly randomized) algorithm, then $r \leftarrow \mathcal{A}(x)$ denotes sampling r from $\mathcal{A}$ on input x over the random coins given to $\mathcal{A}$.*

Notation 4.2. *We use $\Pr[E : D]$ to denote the probability of event E over the distribution of D (sometimes notated as $\Pr_D[E]$). We denote conditional probability of event E conditioned on event C as $\Pr[E|C]$.*

4.1 Probabilistically Checkable Proofs

We first recall the definition of probabilistically checkable proofs. While we give the full definition following the literature on PCPs, we note that the only the syntax will be relevant to our work. We will define the notion of soundness that we will rely on in Section 7.

Definition 4.3 (Probabilistically Checkable Proofs)**.** A probabilistically checkable proof (PCP) of locality ℓ for an NP language $\mathcal{L}$ is a tuple of algorithms $(\mathcal{P}, \mathcal{Q}, \mathcal{V})$ have the following syntax.

- $\mathcal{P}(x, w)$: This algorithm takes as input an instance x and an witness w, and it outputs a proof π of length $L(|x|)$.
- $\mathcal{Q}(1^n, r)$: This algorithm takes as input the input length n, and a uniformly random string r, and output a subset of indices $S \subseteq [L(n)]$ of size ℓ.
- $\mathcal{V}(x, \pi', r)$: The verification algorithm takes as input a string π' of length ℓ, and the random coins r to verify the proof. It either accepts the proof (outputs 1), or rejects the proof (outputs 0).

Moreover, we require the algorithms to satisfy the following properties.

- **Completeness.** For any $(x, w) \in \mathcal{R}_{\mathcal{L}}$, where $\mathcal{R}_{\mathcal{L}}$ is the NP-relation of $\mathcal{L}$, we have
$$\Pr_r[\mathcal{V}(\pi|_Q, r) = 1 : \pi \leftarrow \mathcal{P}(x, w), Q \leftarrow \mathcal{Q}(1^{|x|}, r)] = 1.$$
- **ϵ-Soundness.** For any $x \notin \mathcal{L}$,
$$\Pr_r[\exists \pi^* \in \{0,1\}^{L(|x|)} \text{ s.t. } \mathcal{V}(\pi^*|_Q, r) = 1 : Q \leftarrow \mathcal{Q}(1^{|x|}, r)] \leq \epsilon.$$

In this paper, we will be working with an alternative soundness notion known as (weak) non-signaling soundness (Definition 7.1).

4.2 Hash tree

Definition 4.4 (Hash tree)**.** A hash tree family (HT) consists of the following algorithms:

- $\mathsf{Gen}(1^\lambda) \to \mathsf{hk}$. This probabilistic poly-time algorithm takes as input security parameter λ. It outputs a hash key hk.
- $\mathsf{Hash}(\mathsf{hk}, x) \to \mathsf{rt}$. This deterministic poly-time algorithm takes as input hash key hk and string $x \in \{0,1\}^N$. It outputs a hash root $\mathsf{rt} \in \{0,1\}^{\mathsf{poly}(\lambda, \log N)}$.
- $\mathsf{Open}(\mathsf{hk}, x, i) \to (b, \rho)$. This deterministic poly-time algorithm takes as input hash key hk, string $x \in \{0,1\}^N$ and index $i \in [N]$. It outputs a bit $b \in \{0,1\}$ and an opening $\rho \in \{0,1\}^{\mathsf{poly}(\lambda, \log N)}$.
- $\mathsf{Verify}(\mathsf{hk}, \mathsf{rt}, i, b, \rho)$. This deterministic poly-time algorithm takes as input hash key hk, hash root rt, index $i \in [N]$, bit $b \in \{0,1\}$ and opening $\rho \in \{0,1\}^{\mathsf{poly}(\lambda, \log N)}$. It outputs a bit to signal whether or not the opening is valid.

The scheme is required to satisfy the following properties.

Opening completeness. For any $\lambda \in \mathbb{N}$, any $N = N(\lambda) \leq 2^\lambda$, any $x \in \{0,1\}^N$, and any index $i \in [N]$,

$$\Pr\left[\begin{array}{c} b = x_i \\ \wedge\ \mathsf{Verify}(\mathsf{hk}, \mathsf{rt}, i, b, \rho) = 1 \end{array} : \begin{array}{c} \mathsf{hk} \leftarrow \mathsf{Gen}(1^\lambda), \\ \mathsf{rt} = \mathsf{Hash}(\mathsf{hk}, x), \\ (b, \rho) = \mathsf{Open}(\mathsf{hk}, x, i) \end{array}\right] = 1 - \mathsf{negl}(\lambda).$$

Computational binding w.r.t. opening. For any poly-size adversary $\mathcal{A}$, there exists a negligible function $\mathsf{negl}(\cdot)$ such that for every $\lambda \in \mathbb{N}$,

$$\Pr\left[\begin{array}{c} \mathsf{Verify}(\mathsf{hk}, \mathsf{rt}, i, 0, \rho_0) = 1 \\ \wedge\ \mathsf{Verify}(\mathsf{hk}, \mathsf{rt}, i, 1, \rho_1) = 1 \end{array} : \begin{array}{c} \mathsf{hk} \leftarrow \mathsf{Gen}(1^\lambda), \\ (1^N, \mathsf{rt}, i, \rho_0, \rho_1) \leftarrow \mathcal{A}(\mathsf{hk}) \end{array}\right] = \mathsf{negl}(\lambda).$$

Theorem 4.5 ([Mer88])**.** *Assuming the existence of a collision resistant hash family there exists a hash family with succinct local opening (according to Definition 4.4).*

Remark 4.6 (Opening multiple locations)**.** *One can extend the* Open *algorithm to take a set of indices* $I \subseteq N$*, as opposed to taking a single index* $i \in [N]$*, in the natural way. Namely,*

$$\mathsf{Open}(\mathsf{hk}, x, I) = (\mathsf{Open}(\mathsf{hk}, x, i))_{i \in J}$$

Similarly, one can extend the Verify *algorithm to verify a set of openings, as opposed to a single opening.*

4.3 SNARGs and BARGs

We first recall the notion of a succinct non-interactive argument (SNARG).

Definition 4.7 (SNARG)**.** A succinct non-interactive argument (SNARG) for an NP language $\mathcal{L}$ is a tuple of PPT algorithms $\mathsf{SNARG} = (\mathsf{SNARG.Gen}, \mathsf{SNARG}.\mathcal{P}, \mathsf{SNARG}.\mathcal{V})$ where

- $\mathsf{SNARG.Gen}(1^\lambda, 1^n, 1^m)$: takes as input the security parameter λ, an input length n, a witness length m, and outputs a common reference string crs.

- $\mathsf{SNARG}.\mathcal{P}(\mathsf{crs}, x, w)$: takes as input the common reference string crs, an input $x \in \{0,1\}^n$, and a witness $w \in \{0,1\}^m$, and outputs a proof π.

- $\mathsf{SNARG}.\mathcal{V}(\mathsf{crs}, x, \pi)$: takes as input the common reference string crs, an input $x \in \{0,1\}^n$, and a proof π, and outputs a value in $\{0,1\}$.

The scheme is additionally required to satisfy efficiency and completeness as defined below.

Efficiency: There exists a polynomial p such that on setup parameters $(1^\lambda, 1^n, 1^m)$, the proof has size $|\pi| = p(\lambda, \log n, \log m)$. Moreover, the verifier runtime is $\mathsf{poly}(\lambda, n)$.[9]

Completeness: For all $\lambda \in \mathbb{N}$, all polynomials $n = n(\lambda), m = m(\lambda)$, and all $w \in \{0,1\}^m$ such that $C_\mathcal{L}(x, w) = 1$, where $C_\mathcal{L}$ is the NP verification circuit for $\mathcal{L}$,

$$\Pr\left[\mathsf{SNARG}.\mathcal{V}(\mathsf{crs}, x, \pi) = 1 \;:\; \begin{array}{c} \mathsf{crs} \leftarrow \mathsf{SNARG.Gen}(1^\lambda, 1^n, 1^m), \\ \pi \leftarrow \mathsf{SNARG}.\mathcal{P}(\mathsf{crs}, x, w) \end{array} \right] = 1.$$

A SNARG may satisfy either adaptive or non-adaptive soundness. We define non-adaptive soundness below: For all sufficiently large λ and all poly-size adversaries $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2)$,

$$\Pr\left[\begin{array}{c} x^* \notin \mathcal{L} \\ \wedge\ \mathsf{SNARG}.\mathcal{V}(\mathsf{crs}, x^*, \pi^*) = 1 \end{array} \;:\; \begin{array}{c} (1^n, 1^m, \mathcal{R}, x^*, \mathsf{st}) \leftarrow \mathcal{A}_1(1^\lambda), \\ (\mathsf{crs}) \leftarrow \mathsf{SNARG.Gen}(1^\lambda, 1^n, 1^m, \mathcal{R}), \\ \pi^* \leftarrow \mathcal{A}_2(\mathsf{crs}, x^*, \mathsf{st}) \end{array} \right] \leq \mathsf{negl}(\lambda).$$

We next recall the notion of batch arguments (BARGs).

Definition 4.8 (seBARG for Index-Languages)**.** A (publicly verifiable and non-interactive) somewhere extractable batch argument (seBARG) for index languages is a tuple of PPT algorithms

$$\mathsf{seBARG} = (\mathsf{seBARG.Gen}, \mathsf{seBARG}.\mathcal{P}, \mathsf{seBARG}.\mathcal{V}, \mathsf{seBARG.TDGen}, \mathsf{seBARG.Extract})$$

where

- $\mathsf{seBARG.Gen}(1^\lambda, k, 1^s)$: takes as input the security parameter λ, the number of instances k, and a relation size s, and outputs a common reference string crs.

- $\mathsf{seBARG}.\mathcal{P}(\mathsf{crs}, R, w_1, \ldots, w_k) \to \pi$: takes as input the common reference string crs, the relation R, and witnesses $w_1, \ldots, w_k$, and outputs a proof π.

- $\mathsf{seBARG}.\mathcal{V}(\mathsf{crs}, R, \pi)$: takes as input the common reference string crs, the relation R, and a proof π, and outputs 0 or 1.

- $\mathsf{seBARG.TDGen}(1^\lambda, k, 1^s, i^*)$: takes as input the security parameter λ, the number of instances k, a relation size s, and an index i^* and outputs a common reference string crs and a trapdoor td.

[9]Note that this verifier efficiency can be generically obtained via RAM delegation techniques [CJJ22, KLVW23]. See Remark 2.7 of [WW24].

- $\mathsf{seBARG.Extract}(\mathsf{td}, R, \pi)$: takes as input the trapdoor td, the relation R, a proof π, and outputs a witness w^*.

The scheme is additionally required to satisfy efficiency, completeness, index hiding, and somewhere argument of knowledge as defined below.

Efficiency. There exists a polynomial p such that on setup parameters $(1^\lambda, k, 1^s)$, the crs and proof has size $|\pi| = p(\lambda, \log k, s)$. Moreover, the verifier runtime is $\mathsf{poly}(\lambda, \log k, s)$.

Completeness. For any $\lambda \in \mathbb{N}$, any polynomials $k = k(\lambda), s = s(\lambda)$, any relation R which has size at most s, and any $(w_1, \ldots, w_k)$ such that $R(i, w_i) = 1$ for all $i \in [k]$,

$$\Pr\left[\mathsf{seBARG}.\mathcal{V}(\mathsf{crs}, R, \pi) = 1 \;:\; \begin{array}{c} \mathsf{crs} \leftarrow \mathsf{seBARG.Gen}(1^\lambda, k, 1^s) \\ \pi \leftarrow \mathsf{seBARG}.\mathcal{P}(\mathsf{crs}, R, w_1, \ldots, w_k) \end{array} \right] = 1.$$

Index hiding. For any sufficiently large $\lambda \in \mathbb{N}$, any polynomials $k = k(\lambda), s = s(\lambda)$, any indices $i_0, i_1 \in [k]$, and any poly-size adversary $\mathcal{A}$,

$$\Pr\left[b' = b \;:\; \begin{array}{c} b \leftarrow \{0,1\} \\ (\mathsf{crs}, \mathsf{td}) \leftarrow \mathsf{seBARG.TDGen}(1^\lambda, k, 1^s) \\ b' \leftarrow \mathcal{A}(\mathsf{crs}) \end{array} \right] \leq \frac{1}{2} + \mathsf{negl}(\lambda).$$

Somewhere argument of knowledge. For any sufficintly large $\lambda \in \mathbb{N}$, any polynomials $k = k(\lambda), s = s(\lambda)$, any index $i^* \in [k]$, and any poly-size adversary $\mathcal{A}$,

$$\Pr\left[\begin{array}{c} \mathsf{seBARG}.\mathcal{V}(\mathsf{crs}, R, \pi) = 1 \\ \wedge\, R(i^*, w^*) \neq 1 \end{array} \;:\; \begin{array}{c} (\mathsf{crs}, \mathsf{td}) \leftarrow \mathsf{seBARG.TDGen}(1^\lambda, k, 1^s, i^*) \\ (R, \pi) \leftarrow \mathcal{A}(\mathsf{crs}) \\ w^* \leftarrow \mathsf{seBARG.Extract}(\mathsf{td}, R, \pi) \end{array} \right] \leq \mathsf{negl}(\lambda).$$

Theorem 4.9. *Constructions of* seBARGs *for index relations are known from:*

- *Learning with Error [CJJ21],*
- *Bilinear maps [WW22, KLVW23],*
- *sub-exponential Decisional Diffie-Hellman [CGG$^+$21][10],*
- *sub-exponential Decisional Diffie-Hellman and Quadratic Residuosity [HJKS22].*

4.4 Quasi-Arguments and Hash-and-BARG

We now recall the definition of a quasi-argument [KPY19].

Definition 4.10 (Quasi-argument)**.** A publicly-verifiable non-interactive quasi-argument with non-adaptive non-signaling extraction is a tuple of ppt algorithms $\mathsf{QA} = (\mathsf{QA.Setup}, \mathsf{QA}.\mathcal{P}, \mathsf{QA}.\mathcal{V})$ and a ppt oracle machine QA.LocalGen where

- $\mathsf{QA.Setup}(1^\lambda, 1^s, 1^\ell)$: takes as input the security parameter λ, a circuit size s, and a locality parameter ℓ, and outputs a common reference string crs.

[10] This seBARG construction is only quasi-polynomially secure.

- $\mathsf{QA}.\mathcal{P}(\mathsf{crs}, C, w)$: takes as input the common reference string crs, a circuit C of size s, and an input w, and outputs a proof π and an output y.

- $\mathsf{QA}.\mathcal{V}(\mathsf{crs}, C, \pi, y)$: takes as input the common reference string crs, a circuit C, a proof π, and a claimed output y, and outputs a value in $\{0, 1\}$.

- $\mathsf{QA}.\mathsf{LocalGen}^{\mathcal{A}}(C, T)$: takes as input a circuit C and a subset of wires $T \subseteq [s]$, makes oracle calls to another algorithm $\mathcal{A}$, and outputs a claimed output y, a partial assignment $(\sigma_i)_{i \in T}$ to the wires of C, and auxiliary information aux.

The scheme is required to satisfy completeness and efficiency as defined below.

Completeness. For every $\lambda, s, \ell \in \mathbb{N}$ with $s, \ell \leq 2^\lambda$, and circuit C of size s, input w and output y such that $C(w) = y$, the following holds

$$\Pr\left[\mathsf{QA}.\mathcal{V}(\mathsf{crs}, C, \pi, y) = 1 \ : \ \begin{array}{c} \mathsf{crs} \leftarrow \mathsf{QA}.\mathsf{Setup}(1^\lambda, 1^s, 1^\ell) \\ \pi \leftarrow \mathsf{QA}.\mathcal{P}(\mathsf{crs}, C, w) \end{array} \right] = 1.$$

Efficiency. In the completeness experiment above, all the algorithms run in time $\mathsf{poly}(\lambda, s, \ell)$. Moreover, the size of the proof π is $\mathsf{poly}(\lambda, \ell)$.

The scheme is additionally required to satisfy non-adaptive non-signaling extraction, i.e., the three properties defined below.

Correct distribution. For any $\lambda \in \mathbb{N}$, any polynomials $s = s(\lambda)$ and $\ell = \ell(\lambda)$, any circuit C of size s, any efficient adversary $\mathcal{A}$, and any efficient distinguisher $\mathcal{D}$, we have that

$$\left| \Pr\left[\mathcal{D}(y, \mathsf{aux}) = 1 \ : \ \begin{array}{c} \mathsf{crs} \leftarrow \mathsf{QA}.\mathsf{Setup}(1^\lambda, 1^s, 1^\ell) \\ (y, \pi, \mathsf{aux}) \leftarrow \mathcal{A}(\mathsf{crs}) \\ \text{if } \mathsf{QA}.\mathcal{V}(\mathsf{crs}, C, \pi, y) = 0 : \text{set } y = \mathsf{aux} = \perp \end{array} \right] \right.$$
$$\left. - \Pr\left[\mathcal{D}(y', \mathsf{aux}') = 1 \ : \ (y', \emptyset, \mathsf{aux}') \leftarrow \mathsf{QA}.\mathsf{LocalGen}^{\mathcal{A}}(C, \emptyset) \right] \right| \leq \mathsf{negl}(\lambda).$$

Local consistency. For any $\lambda \in \mathbb{N}$, any polynomials $s = s(\lambda)$ and $\ell = \ell(\lambda)$, any circuit C of size s, any subset $T \subseteq [s]$ where $|T| \leq \ell$, and any efficient adversary $\mathcal{A}$ such that

$$\Pr\left[\begin{array}{c} (\sigma_i)_{i \in T} \text{ locally satisfies } (C, y) \\ \vee\, y = \perp \end{array} \ : \ (y, (\sigma_i)_{i \in T}, \mathsf{aux}) \leftarrow \mathsf{QA}.\mathsf{LocalGen}^{\mathcal{A}}(C, T) \right] \geq 1 - \mathsf{negl}(\lambda)$$

where we say $(\sigma_i)_{i \in T}$ *locally satisfies* (C, y) if

- if $\mathsf{out}_j \in T$, then $\sigma_{\mathsf{out}_j} = y_j$, where we write out_j to denote the j-th output wire of C, and
- $\forall$ gates g in the circuit C s.t. $g_1, g_2, g_{\mathsf{out}} \in T$, we have that $\sigma_{g_{\mathsf{out}}} = g(\sigma_{g_1}, \sigma_{g_2})$, where for a gate g in the circuit C, we write $g_1, g_2, g_{\mathsf{out}}$ to denote the input wires and output wire of g.

Non-signaling. For any $\lambda \in \mathbb{N}$, any polynomials $s = s(\lambda)$ and $\ell = \ell(\lambda)$, any circuit C of size s, any subsets $T_1, T_2 \subseteq [s]$ where $|T_1|, |T_2| \le \ell$, any efficient adversary $\mathcal{A}$, and any efficient distinguisher $\mathcal{D}$, we have that

$$\Big| \Pr\left[\ \mathcal{D}(y_1, (\sigma_i)_{i\in T_1\cap T_2}, \mathsf{aux}_1) = 1 \ :\ (y_1, (\sigma_i)_{i\in[T_1]}, \mathsf{aux}_1) \leftarrow \mathsf{QA.LocalGen}^{\mathcal{A}}(C, T_1)\ \right] \\ -\Pr\left[\ \mathcal{D}(y_2, (\sigma'_i)_{i\in T_1\cap T_2}, \mathsf{aux}_2) = 1 \ :\ (y_2, (\sigma'_i)_{i\in[T_2]}, \mathsf{aux}_2) \leftarrow \mathsf{QA.LocalGen}^{\mathcal{A}}(C, T_2)\ \right] \Big| \le \mathsf{negl}(\lambda).$$

Hash-and-BARG. As observed in several works [BBK+23, JKLV24, JKLM25], the canonical "Hash-and-BARG" protocol of [CJJ22, KVZ21] is indeed a quasi-argument. We formalize this observation as follows.

Theorem 4.11 ([CJJ22, KVZ21])**.** *Assuming the existence of*

- *a hash tree family* HT *(Definition 4.4),*
- *a somewhere-extractable* BARG *family for index-relations (Definition 4.8),*

There exists a quasi-argument scheme as defined in Definition 4.10.

For completeness, we include the full construction and proof in Appendix A.

4.5 Linearly Homomorphic Encryption

The following definition is adapted from [NWW24].

Definition 4.12. A linearly homomorphic encryption scheme with support on $\{0, 1, \dots, n\}$ is defined by a tuple of algorithms

$$\mathsf{LHE} = (\mathsf{LHE.Gen}, \mathsf{LHE.Enc}, \mathsf{LHE.Dec}, \mathsf{LHE.Add})$$

where:

- $\mathsf{LHE.Gen}(1^\lambda, 1^n)$: takes as input the security parameter λ, and a range parameter $n \in N$, and outputs a secret key sk and public key pk.
- $\mathsf{LHE.Enc}(\mathsf{pk}, m)$: takes as input the public key pk and message $m \in \{0, 1, \dots, n\}$, and outputs a ciphertext ct.
- $\mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct})$: takes as input the secret key sk and ciphertext ct, and outputs either some number in $\{0, 1, \dots, n\}$, or $\perp$.
- $\mathsf{LHE.Add}(\mathsf{pk}, \mathsf{ct}_1, \mathsf{ct}_2)$: a *deterministic algorithm* takes as input the public key pk, two ciphertexts ct_1 and ct_2, and outputs a new ciphertext ct.

The scheme is additionally required to satisfy the following properties:

Correctness: For all $\lambda, n \in \mathbb{N}$, and all messages $m \in \{0, 1, \dots, n\}$, it holds that:

$$\Pr\left[\ \mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct}) = m \ :\ \begin{array}{c} (\mathsf{sk}, \mathsf{pk}) \leftarrow \mathsf{LHE.Gen}(1^\lambda, 1^n) \\ \mathsf{ct} \leftarrow \mathsf{LHE.Enc}(\mathsf{pk}, m) \end{array}\ \right] = 1.$$

Evaluation correctness: For all $\lambda, n \in \mathbb{N}$, $(\mathsf{sk}, \mathsf{pk})$ in the support of $\mathsf{LHE.Gen}(1^\lambda, 1^n)$, given ciphertexts ct_1 and ct_2 satisfying that $\mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct}_1) \neq \perp$, $\mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct}_2) \neq \perp$ and $\mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct}_1) + \mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct}_2) \in \{0, 1, \ldots, n\}$, it holds that

$$\mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{LHE.Add}(\mathsf{pk}, \mathsf{ct}_1, \mathsf{ct}_2)) = \mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct}_1) + \mathsf{LHE.Dec}(\mathsf{sk}, \mathsf{ct}_2).$$

Compactness: For all $\lambda, n \in \mathbb{N}$, there exists a polynomial $\mathsf{poly}(\cdot)$ such that $\lambda, n \in \mathbb{N}$ such that the corresponding $|\mathsf{pk}| \leq \mathsf{poly}(\lambda, \log n)$ and $|\mathsf{ct}| \leq \mathsf{poly}(\lambda, \log n)$.

Semantic security: For all $\lambda, n \in \mathbb{N}$, we have:

$$\left| \Pr[\mathsf{Expt}_{\mathcal{A}}^{\mathsf{LHE}}(1^\lambda, 0) = 1] - \Pr[\mathsf{Expt}_{\mathcal{A}}^{\mathsf{LHE}}(1^\lambda, 1) = 1] \right| \leq \mathsf{negl}(\lambda),$$

where for each $b \in \{0, 1\}$, we define $\mathsf{Expt}_{\mathcal{A}}^{\mathsf{LHE}}(1^\lambda, b)$ as follows:

$\mathsf{Expt}_{\mathcal{A}}^{\mathsf{LHE}}(1^\lambda, b)$

1. **Parameters**: $\mathcal{A}$ takes as input 1^λ and outputs an input size 1^n.
2. **Setup**: $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{LHE.Setup}(1^\lambda, 1^n)$, and give pk to $\mathcal{A}$.
3. **Challenge Message Query**:
   (a) $\mathcal{A}$ outputs a challenge message pair (m_0, m_1) where $m_0, m_1 \in \{0, 1, \ldots, n\}$.
   (b) $\mathsf{ct}_b \leftarrow \mathsf{LHE.Enc}(\mathsf{pk}, x_b)$
   (c) Sent ct_b to $\mathcal{A}$.
4. **Experiment Outcome**: $\mathcal{A}$ outputs a bit b' which is the output of the experiment.

Theorem 4.13. *A linearly homomorphic scheme can be constructed assuming any of:*

- *Decisional Diffie-Hellman [ElG84].*
- *Decisional Composite Residuousity [Pai99].*
- *Learning with Errors [Reg05].*

5 The Low-norm Nullstellensatz Hypothesis and the Main Theorem Statement

In this section, we formally state and discuss our low-norm Nullstellensatz hypotheses and then formally state our main theorems.

Definition 5.1 (Nullstellensatz-friendly Code)**.** Let $\mathcal{C} = \{\mathcal{C}_n \subset \{\pm 1\}^{m(n)}\}_n$ denote a family of subsets of $\{\pm 1\}^{m(n)}$ for some polynomial function $m(n)$. Suppose that $\mathcal{C}_n$ is the set of all $x \in \{\pm 1\}^{m(n)}$ satisfying a collection of ℓ-local tests $p_1(x) = 0, \ldots, p_t(x) = 0$. Here, each $p_j(x)$ is a multilinear polynomial that represents a Boolean function.

In this setting, we say that $\mathcal{C}$ is **Nullstellensatz-friendly** if for every polynomial $f(x)$ in the ideal $\mathcal{I} = \langle p_j(x) \rangle_{j\in[t]} = \{f(x) : f(\mathcal{C}) \equiv 0\} \subset \mathbb{R}[x]/\langle x_1^2 - 1, \ldots, x_t^2 - 1\rangle_i$, there exists a decomposition

$$f(x) = \sum_{j=1}^{t} p_j(x) \cdot q_j(x)$$

with $\sum_{j\in[t]} ||q_j||_1 \leq \mathsf{poly}(n) \cdot ||f||_1$.

In other words, $\mathcal{C}$ is Nullstellensatz-friendly if the ideal $\mathcal{I}$ has Nullstellensatz proofs with small relative coefficient size.

Conjecture 5.2 (Low-norm Nullstellensatz hypothesis for the AND code)**.** *The degree two tensor (*And*) code $\mathcal{C} = \{(x, (x_i \wedge x_j)_{i<j})\}_{x\in\{\pm1\}^n}$ is Nullstellensatz-friendly, where $x_i \wedge x_j = \frac{1}{2}(1 + x_i + x_j - x_i x_j)$ computes the Boolean* And *function in the $\{\pm 1\}$ basis.*

Specifically, this code is Nullstellensatz-friendly with respect to the tests $p_{i,j}(x, y) = \frac{1}{2}(1 - y_{i,j} \cdot (x_i \wedge x_j))$.

In general, for any code $\mathcal{C}$, we define the low-norm Nullstellensatz hypothesis for $\mathcal{C}$ to be the hypothesis that $\mathcal{C}$ is Nullstellensatz-friendly.

5.1 On the validity of the low-norm Nullstellensatz hypothesis

While we have not been able to either prove or refute the low-norm Nullstellensatz hypothesis for the AND code (or other non-linear codes), we briefly describe the state of our knowledge about the conjecture.

1. If $\mathcal{C}$ is a *linear* code (more specifically, if the constraint polynomials p_j are parity constraints), then it is Nullstellensatz-friendly.

2. For the case of the AND code, we can show that an important class of polynomials – the "rectangular tensor test polynomials" $p_{S,T}(x, y) = 1/2(1 - y_{S\times T}(x_S \wedge x_T))$ – have Nullstellensatz decompositions with polynomial norm blowup. Here, $(x_S = \prod_{i\in S} x_i, x_T = \prod_{j\in T} x_j)$ are potentially high degree monomials and $y_{S\times T} = \prod_{i\in S, j\in T} y_{i,j}$.

3. By an exhaustive computer search,[11] for $n = 3$, every polynomial $f \in \mathcal{I}$ has a Nullstellensatz decomposition with norm blowup that is no worse than that of a rectangular tensor test.

However, the status of the conjecture remains unclear to us. The **simplest** (in our opinion) sub-problem that we have been unable to resolve is the case of decomposing arbitrary 4-sparse polynomials

$$f(x, y) = 1/4 \cdot (1 - m_1(x, y)) \cdot (1 - m_2(x, y))$$

that are ANDs of monomials (i.e. parities) in the x and y variables, subject to the condition that f vanishes on the AND code. Experimental evidence suggests that these may be the asymptotic worst-case examples in general, making the sub-problem particularly interesting to resolve.

[11] Such a computer search requires quite a bit of care, which we will elaborate on in an updated version of this paper.

5.2 Main Theorem Statement

For any code $\mathcal{C}$ satisfying the low-norm Nullstellensatz hypothesis, our goal is to construct a SNARG for the NP language k-$\mathsf{Lin}(\mathcal{C})$ defined below.

Definition 5.3 (k-$\mathsf{Lin}(\mathcal{C})$)**.** Let $\mathcal{C} = \{\mathcal{C}_n \subset \{\pm 1\}^{m(n)}\}_{n\in\mathbb{N}}$ be a family of codes. We define the computational problem k-$\mathsf{Lin}(\mathcal{C})$ to be the k-Lin problem where solutions are restricted to elements of $\mathcal{C}$.

In other words, a k-$\mathsf{Lin}(\mathcal{C})$ instance consists of a tuple of k-local linear (modulo 2) constraints:

$$w_{i_1} \cdot w_{i_2} \cdot \ldots \cdot w_{i_k} = c \in \{\pm 1\}.$$

A solution $w \in \{\pm 1\}^{m(n)}$ is an element of $\mathcal{C}$ satisfying all of the constraints.

For example, in the case of the AND code, k-$\mathsf{Lin}(\mathcal{C})$ is equivalent to the computational task of solving arbitrary systems of *quadratic equations* modulo 2 with at most k terms per equation, which is NP-complete for $k \geq 2$. Indeed, any "nontrivial" non-linear code $\mathcal{C}$ likely has such an NP-completeness property, although we do not investigate this formally.

Theorem 5.4. *Suppose that there exists a code $\mathcal{C}$ satisfying two conditions:*

- *$\mathcal{C}$ is Nullstellensatz-friendly.*
- *k-$\mathsf{Lin}(\mathcal{C})$ is NP-complete for some constant k.*

Then, assuming the existence of an additively homomorphic encryption scheme and a somewhere extractable BARG *for* NP*, there exist* SNARGs *for* NP*.*

In particular, we have such a construction of SNARGs *for* NP *if Conjecture 5.2 is true.*

6 Local Testing of Hadamard Lifts of Varieties

Let $\mathcal{P} = \{p_1, \ldots, p_t\}$ denote a collection of k-local polynomials in variables $x_1, \ldots, x_n$. In this section, we analyze Sherali-Adams pseudoexpectations in 2^n "extended" $\{\pm 1\}$-variables $\{x_a\}_{a\in\{0,1\}^n}$ satisfying the following conditions:

- $\widetilde{\mathbb{E}}[(x_{a\oplus b} - x_a x_b)^2] \leq \epsilon$ for all $a, b \in \{0,1\}^n$,
- $\widetilde{\mathbb{E}}[p_j(x_1, \ldots, x_n)^2] \leq \epsilon$ for all j,

where we identify $x_i = x_{e_i}$ for the ith standard basis vector $e_i \in \{0,1\}^n$. Informally, we refer to these properties as (approximately) satisfying the linearity tests and $\mathcal{P}$, respectively. Indeed, the pseudoexpectation $\widetilde{\mathbb{E}}$ corresponds to a non-signaling assignment to the variables $\{x_a\}_{a\in\{0,1\}^n}$ that approximately satisfy the linearity tests as well as $\mathcal{P}$.

Let $V = V(\mathcal{P}) \subset \{\pm 1\}^n$ denote the common zero locus of the p_j. Then, the zero locus $\hat{V} \subset \{\pm 1\}^{2^n}$ of $\mathcal{P} \cup \{x_{a\oplus b} - x_a x_b\}_{a,b}$ is precisely the image of V under the Hadamard encoding $\mathsf{Had} : \{0,1\}^n \to \{0,1\}^{2^n}$.

The main result of this section is the following: we prove that if the polynomials $\{p_j\}$ satisfy a *low norm Nullstellensatz* property, defined below, then the following consequence holds:

- For every degree $k+2$ Sherali-Adams pseudoexpectation $\widetilde{\mathbb{E}}$ in the variables $\{x_a\}$ ϵ-satisfying the linearity tests as well as $\mathcal{P}$, and for every polynomial $f \in \mathbb{R}[(x_a)_{a\in\{0,1\}^n}]$ of degree at most k that vanishes on $\hat{V}$, we have that

$$\widetilde{\mathbb{E}}[f] \leq \epsilon \cdot \mathsf{poly}(n) \cdot ||f||_1.$$

To formally state our result, we first define the low norm Nullstellensatz property we make use of. By Nullstellensatz (Theorem 3.23), we know that every polynomial $f(x_1,\ldots,x_n)$ vanishing on V can be written in the form

$$f(x_1,\ldots,x_n) = \sum_j p_j(x)\cdot q_j(x)$$

for some polynomials $q_1,\ldots q_t \in \mathbb{R}[x_1,\ldots,x_n]$, where equality is defined over the ring of multilinear polynomials.

Definition 6.1 (Nullstellsatz coefficient factor)**.** We define the Nullstellensatz coefficient factor $\gamma(\mathcal{P})$ to be

$$\gamma(\mathcal{P}) = \max_{f:||f||_1\leq 1}\ \min_{q_1,\ldots,q_t:f=\sum_j p_j\cdot q_j} \sum_j ||q_j||_1.$$

In other words, $\gamma(\mathcal{P}) \leq \gamma$ if for all f that vanish on V, there exist $q_1,\ldots,q_t$ such that $f = \sum_j p_j \cdot q_j$ and $\sum_j ||q_j||_1 \leq \gamma \cdot ||f||_1$.

In this language, the low-norm Nullstellensatz hypothesis for $\mathcal{C} = V(p_1,\ldots,p_t)$ is exactly the hypothesis that $\gamma(\mathcal{P}) = \mathsf{poly}(n)$.

We are now ready to state the main results of this section.

Lemma 6.2. *Let $\gamma = \gamma(\mathcal{P})$ denote the Nullstellensatz coefficient factor of $\mathcal{P}$. Then, for every degree $k+2$ Sherali-Adams pseudoexpectation $\widetilde{\mathbb{E}}$ in the variables $\{x_a\}$ ϵ-satisfying the linearity tests as well as $\mathcal{P}$, and for every polynomial $f \in \mathbb{R}[(x_a)_{a\in\{0,1\}^{2n}}]$ of degree at most k that vanishes on $\hat{V}$,*

$$\widetilde{\mathbb{E}}[f] \leq O(\epsilon)\cdot\gamma\cdot||f||_1,$$

To prove this result, the key feature of the Hadamard encoding, and the linearity test, that we make use of is that it enables *linearizing* polynomial functions on the Hadamard code.

Definition 6.3 (Linearization)**.** The linear operator $\mathsf{Linearize}$ is defined on N-variate polynomials in the variables $(x_a)_{a\in\{0,1\}^n}$ and maps $\prod_i x_{a_i}$ to $x_{\oplus_i a_i}$.

Indeed, an immediate corollary of Lemma 6.2 is the following.

Corollary 6.4. *Let $\gamma = \gamma(\mathcal{P})$ denote the Nullstellensatz coefficient factor of $\mathcal{P}$. Then, for every degree $k+1$ pseudoexpectation $\widetilde{\mathbb{E}}$ on $\{x_a\}_a$ that ϵ-approximately satisfies the linearity tests as well as $\mathcal{P}$, for every polynomial $f \in \mathbb{R}[x_1,\ldots,x_n]$ that vanishes on V, and for every polynomial $g((x_a)_a)$ of degree at most $k-1$, we have that*

$$\widetilde{\mathbb{E}}[\mathsf{Linearize}(f)\cdot g] \leq O(\epsilon)\cdot\gamma\cdot||\mathsf{Linearize}(f)\cdot g||_1.$$

The corollary follows from the lemma because $\mathsf{Linearize}(f)\cdot g$ always vanishes on $\hat{V}$.

6.1 Proof of Lemma 6.2

To prove Lemma 6.2, we make use of a fact implicitly proved in [CMS18]:

Lemma 6.5. *Let $\widetilde{\mathbb{E}}$ be a degree $k+2$ Sherali-Adams pseudoexpectation that ϵ-approximately satisfies the linearity tests, $\widetilde{\mathbb{E}}$ is $O(\epsilon)$-close to a degree $k+1$ Sherali-Adams pseudoexpectation $\widetilde{\mathbb{E}}'$ that perfectly satisfies the linearity tests.*

As a result, we will now instead assume that $\widetilde{\mathbb{E}}$ is a degree $k+1$ pseudoexpectation that perfectly satisfies the linearity tests. In this situation, we make use of the following claim.

Claim 6.6. *If $f, g \in \mathbb{R}[(x_a)_a]$ have degree at most $k+1$ and $\mathsf{Linearize}(f) = \mathsf{Linearize}(g)$, then $\widetilde{\mathbb{E}}[f] = \widetilde{\mathbb{E}}[g]$.*

To prove this claim, we simply prove that $\widetilde{\mathbb{E}}[f] = \widetilde{\mathbb{E}}[\mathsf{Linearize}(f)]$. This holds because of the following simple lemma.

Lemma 6.7. *If f has at most degree k, then $f - \mathsf{Linearize}(f) = \sum_{a,b} g_{a,b} \cdot p_{a,b}$ for some polynomials $g_{a,b}$ of degree at most $k-2$.*

Proof. Since $\mathsf{Linearize}(f) + \mathsf{Linearize}(g) = \mathsf{Linearize}(f+g)$, it suffices to prove the lemma when f is a monomial $f(x) = \prod_{i=1}^{k} x_{a^{(i)}}$. In this case, we know that $\mathsf{Linearize}(f) = x_{a^*}$ for $a^* = \oplus_i a^{(i)}$. We now observe that

$$f - \mathsf{Linearize}(f) = \sum_{i=1}^{k} (x_{a^{(1)} \oplus \ldots \oplus a^{(i)}} \cdot x_{a^{(i+1)}} - x_{a^{(1)} \oplus \ldots \oplus a^{(i+1)}}) \cdot x_{a^{(i+2)}} \cdot \ldots \cdot x_{a^{(k)}}. \qquad \square$$

Finally, since $f - \mathsf{Linearize}(f) = \sum_{a,b} p_{a,b} \cdot g_{a,b}$ as above, we conclude that $\widetilde{\mathbb{E}}[f - \mathsf{Linearize}(f)] = \sum_{a,b} \widetilde{\mathbb{E}}[p_{a,b} \cdot g_{a,b}] = 0$.

Using this claim, we are now ready to prove Lemma 6.2. Let $\widetilde{\mathbb{E}}$ be a pseudoexpectation as in Lemma 6.2, and let $f((x_a)_a)$ be a degree k polynomial vanishing on $\widehat{V}$. Let $\tilde{f}(x_1, \ldots, x_n) = f((x_a = \prod_i x_i^{a_i}))$ be f's corresponding high degree polynomial in the variables $x_1, \ldots, x_n$, and observe that $\mathsf{Linearize}(f) = \mathsf{Linearize}(\tilde{f})$. Since f vanishes on $\widehat{V}$, it follows that $\tilde{f}$ vanishes on V. Thus, by our hypothesis, there exist polynomials $q_1, \ldots, q_t$ such that

$$\tilde{f} = \sum_j p_j \cdot q_j$$

and $\sum_j ||q_j||_1 \le \gamma \cdot ||\tilde{f}||_1 \le \gamma \cdot ||f||_1$. Now, we observe that

$$\mathsf{Linearize}(f) = \sum_j \mathsf{Linearize}(p_j \cdot q_j) = \sum_j \mathsf{Linearize}(p_j \cdot \mathsf{Linearize}(q_j)),$$

which implies that

$$\widetilde{\mathbb{E}}[f] = \sum_j \widetilde{\mathbb{E}}[p_j \cdot \mathsf{Linearize}(q_j)] \le \epsilon \cdot \sum_j ||\mathsf{Linearize}(q_j)||_1 \le \epsilon \cdot \sum_j ||q_j||_1 \le \epsilon \cdot \gamma \cdot ||f||_1.$$

This completes the proof of Lemma 6.2.

7 Weak Soundness of Hadamard-Encoded nsPCPs

In this section, we complete the proof of weak soundness of the Hadamard-based nsPCP for k-$\mathsf{Lin}(\mathcal{C})$, where $\mathcal{C}$ is a code satisfying the low-norm Nullstellensatz hypothesis.

Definition 7.1 (PCP with weak non-signaling soundness)**.** We say that a PCP scheme $(\mathcal{P}, \mathcal{Q}, \mathcal{V})$ (as defined in Definition 4.3) satisfies weak ℓ-non-signaling soundness if there exists a polynomial $\mathsf{poly}(\cdot)$ such that for every instance not in L, and every family of non-signaling distributions $\{\mathcal{D}_S\}_{|S|\leq \ell}$, there exists some set S with $|S| \leq \ell$ such that for instances of length n,

$$\Pr\left[\ \mathcal{V}(\pi) = 1\ \ :\ \ \pi \leftarrow \mathcal{D}_S\ \right] \leq 1 - \frac{1}{\mathsf{poly}(n)}.$$

Equivalently (by Claim 3.18), this states that for all degree ℓ Sherali-Adams pseudoexpectations $\widetilde{\mathbb{E}}$ over variables x_i corresponding to the bits of the PCP string π, there exists a PCP test polynomial p_r — the polynomial $p_r(x)$ computes the Boolean function which is 0 if and only if the verifier accepts when using random coins r — such that $\widetilde{\mathbb{E}}[p_r^2] \geq \frac{1}{\mathsf{poly}(n)}$.

Consider a code $\mathcal{C} \subseteq \{\pm 1\}^m$ defined by the ℓ-local tests $p_1(x) = 0, \ldots, p_t(x) = 0$. We now define a variant of the Hadamard PCP for the language k-$\mathsf{Lin}(\mathcal{C})$. Recall that an instance $x = ((T_1, c_1), \ldots, (T_t, c_t))$ of k-$\mathsf{Lin}(\mathcal{C})$ is a tuple of k-local linear constraints. Each constraint is represented by a subset of indices $T \subseteq \{0,1\}^m$ and a value $c \in \{\pm 1\}$. A string w is a valid witness for instance x if it satisfies all of the constraints.

Construction 7.2 (Hadamard PCP for k-$\mathsf{Lin}(\mathcal{C})$)**.** *We define the* PCP *via the tuple* $(\mathcal{P}, \mathcal{Q}, \mathcal{V})$ *as follows. Let* $L = k + \ell + O(1)$.

- $\mathcal{P}(x, w)$: *For instance* $x = ((T_1, c_1), \ldots, (T_t, c_t))$ *and witness* $w \in \{\pm 1\}^m$*, output the string*

$$\pi = \left\{ \prod_{i \in [m] : a_i = 1} w_i \right\}_{a \in \{0,1\}^m} \in \{\pm 1\}^{2^m}$$

- $\mathcal{Q}(r \leftarrow \{0,1\}^{L \cdot m})$: *Parse the randomness as* $a_1, \ldots, a_L \in \{0,1\}^m$*, and output* $S = \{q_1, \ldots, q_L\}$.
- $\mathcal{V}(x, S, \pi|_S)$: *Given the set* S*, the verifier conducts the following tests.*
  - ***Linearity test:*** *If* Q *contains the strings* $a, b, a \oplus b$*, then verify that*

  $$\pi_a \cdot \pi_b \cdot \pi_{a \oplus b} = 1.$$

  - k-Lin ***tests:*** *Let* e_j *denote the indicator vector which is 1 at index* j *and 0 everywhere else. If there exists some constraint set* $T_i = \{t_1, \ldots, t_{k'}\}$ *such that* $S \supset \{e_{t_1}, \ldots, e_{t_{k'}}\}$*, then verify that*

  $$\pi_{e_{t_1}} \cdot \pi_{e_{t_2}} \cdot \cdots \cdot \pi_{e_{t_{k'}}} = c_i.$$

  - ***Code tests:*** *If there exists some code test* p_j *on indices* $\{t_1, \ldots, t_\ell\}$ *such that* S *contains* $\{e_{t_1}, \ldots, e_{t_\ell}\}$*, then verify that*

  $$p_j(\pi_{e_{t_1}}, \ldots, \pi_{e_{t_\ell}}) = 0.$$

In the rest of this section, we prove the following.

Lemma 7.3. *Suppose that $\mathcal{C}$ satisfies the low-norm Nullstellensatz hypothesis. Then, the Hadamard* PCP *for k-*Lin*$(\mathcal{C})$ satisfies weak non-signaling soundness for a sufficiently large locality $L = k + \ell + O(1)$.*

We proceed to prove Lemma 7.3 by contradiction. Fix a false instance $(T_1, c_1, \ldots, T_t, c_t)$ of k-Lin$(\mathcal{C})$, and suppose that a Sherali-Adams pseudoexpectation $\widetilde{\mathbb{E}}$ passes every linearity test, code test and k-Lin test with error at most ϵ. Again using Lemma 6.5, we may assume that the linearity tests pass with 0 error (up to a constant factor loss in ϵ). Furthermore, we may invoke Lemma 6.2, which states that for any polynomial $p(x_1, \ldots, x_m)$ that vanishes on $\mathcal{C}$, we have that

$$\widetilde{\mathbb{E}}[\mathsf{Linearize}(p)] \le \epsilon \cdot ||p||_1 \cdot \mathsf{poly}(n)$$

for a fixed polynomial $\mathsf{poly}(\cdot)$ depending on the low-norm Nullstellsatz hypothesis. We will use this property to derive a contradiction.

7.1 The k-Lin polynomial

Let $G = \{g_1, \ldots, g_t\}$ denote the set of all k-Lin tests, meaning that

$$g_j(x) = \frac{1}{2}\Big(1 - c_j \cdot \prod_{i \in T_j} x_i\Big) = \frac{1 - c_j \cdot x_{T_j}}{2}.$$

Based on these tests, we define a key polynomial associated with the instance,

$$p^*(x) = \frac{1}{2^t} \sum_{U \subset [t]} \frac{1 - \prod_{i \in U} c_i \cdot x_{T_i}}{2}.$$

For any fixed assignment $x \in \{\pm 1\}^m$, $p^*(x)$ is equal to the probability, over a random subset of k-Lin tests, that an odd number of the selected k-Lin tests fail. In other words, letting $g(x) = (g_1(x), g_2(x), \ldots, g_t(x))$ denote the vector in $\{0,1\}^t$ recording which k-Lin tests failed, $p^*(x)$ is the expected $\mathbb{F}_2$-inner product of $g(x)$ with a random vector in $\mathbb{F}_2^t$.

Therefore, because our k-Lin instance is assumed unsatisfiable, we have that for every every $x \in \mathcal{C}$,

$$p^*(x) = 1/2.$$

Thus, we can conclude that

$$\widetilde{\mathbb{E}}[\mathsf{Linearize}(p^*) - 1/2] = \epsilon \cdot \mathsf{poly}(n) \cdot ||p^*||_1 = \epsilon \cdot \mathsf{poly}(n),$$

as $||p^*||_1 \le 3/2 = O(1)$.

On the other hand, we also claim the following.

Claim 7.4. *If $\widetilde{\mathbb{E}}$ passes all k-*Lin *tests with error $\le \epsilon$, then $\widetilde{\mathbb{E}}[\mathsf{Linearize}(p^*)] = O(t \cdot \epsilon)$.*

But the equation $1/2 + \epsilon \cdot \mathsf{poly}(n) = O(t \cdot \epsilon)$ is a contradiction for sufficiently small (inverse polynomial) ϵ, proving Lemma 7.3. Thus, it suffices to prove the claim.

Proof of claim. We define a sequence of hybrid polynomials p_j^* for $0 \le j \le t$:

$$p_j^*(x) = \frac{1}{2^{t-j}} \sum_{U \subset [t-j]} \frac{1 - \prod_{i \in T} c_i \cdot x_{T_i}}{2}.$$

In other words, p_j^* is the polynomial analogous to p^* associated with the first $t - j$ k-Lin tests $g_1, \ldots, g_{t-j}$. Note that $p_0^* = p^*$ and $p_t^* = 0$ is the constant zero polynomial. Thus, it suffices to show that

$$\widetilde{\mathbb{E}}[\mathsf{Linearize}(p_j^* - p_{j+1}^*)] = O(\epsilon)$$

for every j.

To see this, letting $Q_U(x) = \prod_{i \in U} c_i \cdot x_{T_i}$, we write

$$\begin{aligned}
p_j^*(x) - p_{j+1}^*(x) &= \frac{1}{2^{t-j-1}} \sum_{U \subset [t-j-1]} \frac{1}{2}\Big(\frac{1 - c_{t-j} \cdot x_{T_{t-j}} \cdot \prod_{i \in U} c_i \cdot x_{T_i}}{2} - \frac{1 - \prod_{i \in U} c_i \cdot x_{T_i}}{2}\Big) \\
&= \frac{1}{2^{t-j-1}} \sum_{U \subset [t-j-1]} \frac{1}{2}\Big(\frac{1 - c_{t-j} \cdot x_{T_{t-j}} \cdot Q_U(x)}{2} - \frac{1 - Q_U(x)}{2}\Big) \\
&= \frac{1}{2^{t-j-1}} \sum_{U \subset [t-j-1]} \frac{Q_U(x)}{2}\Big(\frac{1 - c_{t-j} \cdot x_{T_{t-j}}}{2}\Big) \\
&= g_{t-j}(x) \cdot \frac{1}{2^{t-j-1}} \sum_{U \subset [t-j-1]} \frac{Q_U(x)}{2}.
\end{aligned}$$

Thus, by Claim 6.6 we conclude that

$$\begin{aligned}
\widetilde{\mathbb{E}}[p_j^* - p_{j+1}^*] &= \widetilde{\mathbb{E}}\Big[g_{t-j} \cdot \mathsf{Linearize}\Big(\frac{1}{2^{t-j-1}} \sum_{U \subset [t-j-1]} \frac{Q_U(x)}{2}\Big)\Big] \\
&\le \epsilon \cdot \Big|\Big|\frac{1}{2^{t-j-1}} \sum_{U \subset [t-j-1]} \frac{Q_U(x)}{2}\Big|\Big|_1 \\
&= O(\epsilon),
\end{aligned}$$

because $||Q_U(x)||_1 = 1$ for all U. This completes the proof. $\square$

8 Non-Adaptive SNARG for NP via nsPCPs

In this section, we construct a SNARG for NP using the following ingredients:

- A quasi-argument $\mathsf{QA} = (\mathsf{QA.Setup}, \mathsf{QA}.\mathcal{P}, \mathsf{QA}.\mathcal{V}, \mathsf{QA.LocalGen})$.
- A linearly homomorphic encryption scheme $\mathsf{LHE} = (\mathsf{LHE.Gen}, \mathsf{LHE.Enc}, \mathsf{LHE.Add}, \mathsf{LHE.Dec})$.
- A polynomial-length $\mathsf{PCP} = (\mathsf{PCP}.\mathcal{P}.\mathsf{PCP}.\mathcal{Q}, \mathsf{PCP}.\mathcal{V})$ with constant locality, which is weakly non-signaling sound when Hadamard-encoded.

For simplicity, we assume that the instance x is encoded as a description of the PCP tests.[12] In particular, we parse $x = ((Q_1, v_1), \ldots, (Q_t, v_t))$. If Γ is a PCP for x, the PCP verifier checks that $v_i(\Gamma_{Q_i}) = 1$ for $i \leftarrow [t]$, where Γ_{Q_i} denotes the projection of Γ onto the bits in Q_i.

8.1 Construction

We are now ready to define our construction $\mathsf{SNARG} = (\mathsf{SNARG.Gen}, \mathsf{SNARG}.\mathcal{P}, \mathsf{SNARG}.\mathcal{V})$ for an NP language $\mathcal{L}$. Let $M(n, m)$ be the length of the PCP output by $\mathsf{PCP}.\mathcal{P}$ on an instance of length n and witness of length m. Let $K = k + 2$, where $k = O(1)$ is the maximum size of a query set Q for inputs of length n. Let P be the maximum size of a verifier function v for inputs of length n.

$\underline{\mathsf{SNARG.TDGen}(1^\lambda, 1^n, 1^m, \mathcal{Q} = \{q_1, \ldots, q_K\} \subseteq \{0,1\}^{M(n,m)}) \to (\mathsf{crs}, \mathsf{td}).}$

1. Let s be the size of the circuit $\widehat{\mathcal{C}}$ defined in Step (2) of the prover algorithm below.
2. For $i \in [K]$ sample $(\mathsf{sk}_i, \mathsf{pk}_i) \leftarrow \mathsf{LHE.Gen}(1^\lambda, 1^M)$ and for $j \in [M]$ compute
$$\mathsf{ct}_{i,j} = \mathsf{LHE.Enc}(\mathsf{pk}_i, q_{i,j}).$$
Let $\overrightarrow{\mathsf{pk}} = (\mathsf{pk}_1, \ldots, \mathsf{pk}_K)$ and $\overrightarrow{\mathsf{ct}} = \{\mathsf{ct}_{i,j}\}_{i \in [K], j \in [M]}$.
3. Let $\ell = \mathsf{poly}(\lambda, K, P, |\mathsf{LHE.Add}(\mathsf{pk}_1, \cdot, \cdot)|)$.
4. Sample a quasi-argument common reference string $\mathsf{crs}_{\mathsf{QA}} \leftarrow \mathsf{QA.Setup}(1^\lambda, 1^s, 1^\ell)$.
5. Output $\mathsf{crs} = (\mathsf{crs}_{\mathsf{QA}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}})$, and $\mathsf{td} = \{\mathsf{sk}_i\}_{i \in [K]}$.

$\underline{\mathsf{SNARG.Gen}(1^\lambda, 1^n, 1^m) \to \mathsf{crs}.}$

1. Let $(\mathsf{crs}, \mathsf{td}) \leftarrow \mathsf{SNARG.TDGen}(1^\lambda, 1^n, 1^m, 0^{K \cdot M})$.
2. Output crs.

$\underline{\mathsf{SNARG}.\mathcal{P}(\mathsf{crs}, x, w) \to \pi.}$

1. Parse $\mathsf{crs} = (\mathsf{crs}_{\mathsf{QA}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}})$, $\overrightarrow{\mathsf{pk}} = (\mathsf{pk}_1, \ldots, \mathsf{pk}_K)$, $\overrightarrow{\mathsf{ct}} = \{\mathsf{ct}_{i,j}\}_{i \in [K], j \in [M]}$, and $x = ((Q_1, v_1), \ldots, (Q_t, v_t))$.

[12] If we compose our construction with a SNARG for P which maps the real instance to this encoded version, we can easily remove this assumption.

2. We define a computation $\widehat{\mathcal{C}}[x, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}]$ on an input $w \in \{0,1\}^m$ as follows:

   (a) **Generate PCP string.** Compute $\Gamma \leftarrow \mathsf{PCP}.\mathcal{P}(x, w)$.

   (b) **Check PCP tests.** For $j \in [t]$: compute

   $$\rho_j = \begin{cases} 1 & \text{if } v_j(\Gamma_{Q_j}) = 0 \\ 0 & \text{otherwise.} \end{cases}$$

   Compute $\mathsf{out} = \bigwedge_{j \in [t]} \rho_j$ in a binary tree.

   (c) **Homomorphically evaluate PCP queries.**[13]
   For each $i \in [K]$, do the following computation:

      i. For all $j \in [M]$, let

      $$\alpha_{i,j} = \begin{cases} \mathsf{ct}_{i,j} & \text{if } \Gamma_j = 1 \\ 0 & \text{if } \Gamma_j = 0. \end{cases}$$

      In the following, we abuse the notation $\mathsf{LHE.Add}(\mathsf{pk}, \mathsf{ct}, 0) = \mathsf{LHE.Add}(\mathsf{pk}, 0, \mathsf{ct}) = \mathsf{ct}$.

      ii. Set $\beta_{i,1} \leftarrow \alpha_{i,1}$.

      iii. For $j = 2, \ldots, M$: Compute $\beta_{i,j} \leftarrow \mathsf{LHE.Add}(\mathsf{pk}_i, \beta_{i,j-1}, \alpha_{i,j})$.

      iv. Set $\mathsf{ans}_i = \beta_{i,M}$.

   (d) Output $(\mathsf{out}, \{\mathsf{ans}_i\}_{i \in [K]})$.

3. Compute a quasi-argument proof $(\pi_{\mathsf{QA}}, (\mathsf{out}, \{\mathsf{ans}_i\}_{i\in[K]})) \leftarrow \mathsf{QA}.\mathcal{P}(\mathsf{crs}_{\mathsf{QA}}, \widehat{\mathcal{C}}[x, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}], w)$.

4. Output $\pi := (\pi_{\mathsf{QA}}, (\mathsf{out}, \{\mathsf{ans}_i\}_{i\in[K]}))$.

$\mathsf{SNARG}.\mathcal{V}(\mathsf{crs}, x, \pi) \rightarrow 0/1.$

1. Parse $\mathsf{crs} = (\mathsf{crs}_{\mathsf{QA}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}})$ and $\pi = (\pi_{\mathsf{QA}}, \mathsf{out}, (\{\mathsf{ans}_i\}_{i\in[K]}))$.

2. Compute $\widehat{\mathcal{C}}[x, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}]$ as in Step (2) of the prover algorithm $\mathcal{P}$.

3. Accept if $\mathsf{out} = 1$ and $\mathsf{QA}.\mathcal{V}(\mathsf{crs}_{\mathsf{QA}}, \widehat{\mathcal{C}}[x, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}], \pi_{\mathsf{QA}}, (\mathsf{out}, \{\mathsf{ans}_i\}_{i\in[K]})) = 1$.

Efficiency of the above construction follows from efficiency of QA and LHE, and the fact that K and P are $O(1)$ (so the quasi-argument proof is succinct). Completeness follows from the completeness of QA and correctness of LHE.

8.2 Soundness Analysis

In this section, we will prove the following theorem.

Theorem 8.1. *Suppose there exists a polynomial-sized prover $\mathcal{A}$ such that for some $x^* \in \{0,1\}^n$ (possibly in the language),*

$$\Pr\left[\mathsf{SNARG}.\mathcal{V}(\mathsf{crs}, x^*, \pi) = 1 \ : \ \begin{array}{c} \mathsf{crs} \leftarrow \mathsf{SNARG.Gen}(1^\lambda, 1^n, 1^m), \\ \pi \leftarrow \mathcal{A}(\mathsf{crs}) \end{array} \right] \geq \epsilon(\lambda),$$

[13] In the actual scheme, when $\mathsf{crs} \leftarrow \mathsf{SNARG.Gen}(1^\lambda, 1^n, 1^m)$, the queries are all 0.

for some non-negligible function ϵ. Then, there exists a degree K Sherali-Adams pseudoexpectation $\widetilde{\mathbb{E}}$ defined on the Hadamard-extended variables $\{x_S, y_T\}_{S\subseteq[m],T\subseteq[\rho(m)]}$ such that for Hadamard-encoded nsPCP *test polynomials $p(x,y)$, we have $\widetilde{\mathbb{E}}[p(x,y)^2] \le \mathsf{negl}(\lambda)$.*

Note that when combined with Lemmas 6.2 and 7.3, this implies soundness of the SNARG when instantiated with the Hadamard-encoded nsPCP for k-$\mathsf{LIN}(\mathcal{C})$. We proceed in multiple steps as follows.

- In Section 8.2.1, we show how to construct a local assignment generator which interacts with a cheating prover to extract wire values from the computation of $\widehat{\mathcal{C}}$ and the evaluated (encrypted) answers to the Hadamard-encoded PCP queries.
- In Section 8.2.2, we show that these answers, when decrypted, are a consistent non-signaling strategy for the Hadamard-encoded PCP.
- In Section 8.2.3, we show that this implies a degree K Sherali-Adams pseudoexpectation which satisfies the linearity and PCP tests.

8.2.1 From a SNARG Cheating Prover to a Local Assignment Generator

Consider the following extractor algorithm LocalGen (Fig. 1) that on receives an instance x^*, a subset of wires T of size at most ℓ, and an *ordered* set $\mathcal{Q}$ of K queries $q_1, \ldots, q_K \in \{0,1\}^M$ and does the following:

$\underline{\mathsf{LocalGen}^{\mathcal{A}}(x^*, T, \mathcal{Q} = (q_1, \ldots, q_K)):}$

- Repeat the following experiment at most $\mathsf{poly}(\lambda, 1/\epsilon)$ times:
  - Call $(\mathsf{crs} = (\mathsf{crs}'_{\mathsf{QA}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}), \mathsf{td} = \overrightarrow{\mathsf{sk}}) \leftarrow \mathsf{SNARG.TDGen}(1^\lambda, 1^n, 1^m, \mathcal{Q})$.
  - Let $\widehat{\mathcal{C}}[x^*, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}]$ be the circuit defined in Step (2) of the $\mathsf{SNARG}.\mathcal{P}$ algorithm.
  - Interact with $\mathsf{QA.LocalGen}(\widehat{\mathcal{C}}[x^*, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}], T)$: Whenever QA.LocalGen sends a query $\mathsf{crs}_{\mathsf{QA}}$, send $\pi = (\pi_{\mathsf{QA}}, (\mathsf{out}, \overrightarrow{\mathsf{ans}})) \leftarrow \mathcal{A}(\mathsf{crs}_{\mathsf{QA}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}})$ along with $\mathsf{aux} = (\overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}})$ back to QA.LocalGen.
  - If QA.LocalGen outputs $(\perp, \perp)$, skip to the next experiment. Otherwise, obtain output $((\sigma_j)_{j\in T}, (\mathsf{out}, (\mathsf{ans}_i)_{i\in[K]}), \mathsf{aux})$ and exit the loop.
- Output $((\sigma_j)_{j\in T}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}})$.
- If none of the experiments succeed, output $\perp$.

Figure 1: *Local assignment generator for* SNARG.

Claim 8.2. $\mathsf{LocalGen}^{\mathcal{A}}$ *satisfies the following properties:*

Success probability. *On every input $(x^*, T, \mathcal{Q})$,* $\mathsf{LocalGen}^{\mathcal{A}}$ *outputs $((\sigma_j)_{j\in T}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}})$ where $\mathsf{out} = 1$ and does not output $\perp$ except with negligible probability.*

Local consistency. *For* $((\sigma_j)_{j\in T}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T, \mathcal{Q})$*, we have that* $(\sigma_j)_{j\in T}$ *locally satisfies* $(\widehat{\mathcal{C}}[x^*, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}], (\mathsf{out}, \overrightarrow{\mathsf{ans}}))$.

Non-signaling. *For any subsets* $T_1, T_2 \subseteq [s]$ *where* $|T_1|, |T_2| \leq \ell$*, any efficient adversary* $\mathcal{A}$*, and any efficient distinguisher* $\mathcal{D}$*, we have that*

$$\Bigg| \Pr\left[\mathcal{D}((\sigma_j)_{j\in T_1\cap T_2}, \mathsf{aux}) = 1 \ : \ \begin{array}{c} ((\sigma_j)_{j\in T_1}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_1, \mathcal{Q}), \\ \mathsf{aux} = (\mathsf{out}, \overrightarrow{\mathsf{ans}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \end{array} \right]$$
$$- \Pr\left[\mathcal{D}((\sigma'_j)_{j\in T_1\cap T_2}, \mathsf{aux}') = 1 \ : \ \begin{array}{c} ((\sigma'_j)_{j\in T_2}, (\mathsf{out}', \overrightarrow{\mathsf{ans}}'), \overrightarrow{\mathsf{pk}}', \overrightarrow{\mathsf{sk}}', \overrightarrow{\mathsf{ct}}') \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_2, \mathcal{Q}) \\ \mathsf{aux}' = (\mathsf{out}', \overrightarrow{\mathsf{ans}}', \overrightarrow{\mathsf{pk}}', \overrightarrow{\mathsf{sk}}', \overrightarrow{\mathsf{ct}}') \end{array} \right] \Bigg|$$
$$\leq \mathsf{negl}(\lambda).$$

Proof. We show each property by relying on semantic security of the LHE scheme, and properties of the QA scheme.

Success probability. To show that $\mathsf{LocalGen}^{\mathcal{A}}$ outputs $\perp$ with $\mathsf{negl}(\lambda)$, it suffices to show that that each run of the $\mathsf{LocalGen}^{\mathcal{A}}$ succeeds with probability $\mathsf{poly}(\epsilon(\lambda) - \mu(\lambda))$, where μ is some negligible function.

We show this in two steps. First, by the semantic security of LHE, it is easy to see that the output of $\mathsf{SNARG.Gen}(1^\lambda, 1^n, 1^m)$ and $\mathsf{SNARG.TDGen}(1^\lambda, 1^n, 1^m, \mathcal{Q})$ are indistinguishable. Therefore, there exists a negligible function $\mu_1(\lambda)$ such that $\mathcal{A}$ still creates accepting proofs (where $\mathsf{out} = 1$) with respect to $\mathsf{crs} \leftarrow \mathsf{SNARG.TDGen}(1^\lambda, 1^n, 1^m, \mathcal{Q})$ with probability $\epsilon_1(\lambda) = \epsilon(\lambda) - \mu_1(\lambda)$.

Now, consider $(\mathsf{crs}_{\mathsf{QA}}, \mathsf{pk}, \mathsf{ct})$ in the support of $\mathsf{SNARG.TDGen}(1^\lambda, 1^n, 1^m, \mathcal{Q})$.

Define

$$\delta_{\overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}} = \Pr\left[\begin{array}{c} \pi \leftarrow \mathcal{A}(\mathsf{crs}) \\ \wedge\ \mathsf{SNARG}.\mathcal{V}(\mathsf{crs}, x^*\pi) = 1 \end{array} \ : \ \begin{array}{c} \mathsf{crs} \leftarrow \mathsf{SNARG.TDGen}(1^\lambda, 1^n, 1^m, \mathcal{Q}) \\ \wedge\ (\mathsf{crs}_{\mathsf{QA}}, \overrightarrow{\mathsf{pk}}', \overrightarrow{\mathsf{ct}}') \leftarrow \mathsf{crs} \\ \wedge\ \overrightarrow{\mathsf{pk}} = \overrightarrow{\mathsf{pk}}' \\ \wedge\ \overrightarrow{\mathsf{ct}} = \overrightarrow{\mathsf{ct}}' \end{array} \right],$$

i.e. the probability that $\mathcal{A}$ creates accepting proofs sampling the crs on the public key $\overrightarrow{\mathsf{pk}}$ and ciphertexts $\overrightarrow{\mathsf{ct}}$. We say $\overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}$ is "good" if $\delta_{\overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}} \geq \epsilon_1/2$. By a standard counting argument, since $\mathcal{A}$ creates accepting proofs with probability ϵ_1, we have that

$$\Pr\left[\overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}} \text{ is good } \ : \ \begin{array}{c} \mathsf{crs} \leftarrow \mathsf{SNARG.TDGen}\mathcal{Q}(1^\lambda, 1^n, 1^m, \mathcal{Q}) \\ \wedge\ (\mathsf{crs}_{\mathsf{QA}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{crs} \end{array} \right] \geq \epsilon_1/2.$$

Now, fix a "good" tuple $(\overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}})$. By the correct distribution and non-signaling properties, we have that $\mathsf{QA.LocalGen}^{\mathcal{A}}(\widehat{\mathcal{C}}[x^*, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}], T)$ doesn't output $(\perp, \perp)$ with probability at least $\delta_{\overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}} - \mu_2(\lambda)$, where $\mu_2(\lambda)$ is some non-negligible function.

Therefore, in a single run of the experiment, the probability that $\mathsf{LocalGen}^{\mathcal{A}}$ outputs a tuple where $\mathsf{out} = 1$ and does not output $\perp$, is at least $\epsilon_1/2 \cdot (\epsilon_1/2 - \mu_2(\lambda)) = \mathsf{poly}(\epsilon - \mu(\lambda))$, where μ is some negligible function. Therefore, by repeating the experiment $\mathsf{poly}(1/\epsilon, \lambda)$ times, one can ensure that $\mathsf{LocalGen}^{\mathcal{A}}$ succeeds with probability at least $1 - \mathsf{negl}(\lambda)$, as desired.

Local consistency. If $((\sigma_i)_{i\in T}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \mathsf{aux}) \leftarrow \mathsf{QA.LocalGen}^{\mathcal{A}}(\widehat{\mathcal{C}}[x^*, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}]$ (i.e. does not output $\perp$), by the local consistency of QA.LocalGen, $(\sigma_i)_{i\in T}$ locally satisfies $(\hat{C}[x^*, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}], (\mathsf{out}, \overrightarrow{\mathsf{ans}}))$ with probability $1 - \mathsf{negl}(\lambda)$.

Non-Signaling. This follows straightforwardly from the non-signaling property of QA.

□

8.2.2 To a Consistent Non-Signaling Strategy that Satisfies Tests

In this section, we treat all query sets $\mathcal{Q} \subseteq \{0,1\}^M$ as *ordered*.

Consider the following algorithm.

> $\mathsf{nsPCP\text{-}answers}(\mathcal{Q})$:
>
> - Call $\{(\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, \varnothing, \mathcal{Q})$. (If $\mathsf{LocalGen}^{\mathcal{A}}$ fails, output $\perp$).
> - For $i \in [K]$, compute $b_i = \mathsf{LHE.Dec}(\mathsf{sk}_i, \mathsf{ans}_i) \pmod 2$.
> - Output $(b_1, \ldots, b_k)$.

We will now prove that the answers provided by the above algorithm constitute a consistent non-signaling strategy for the Hadamard-encoded nsPCP, i.e., satisfy the following properties:

- **Linear consistency:** If $\mathcal{Q}$ contains q_u, q_v, q_w such that $q_u \oplus q_v \oplus q_w = 0^M$, then $(b_1, \ldots, b_K)$ satisfies that $b_u \oplus b_v \oplus b_w = 0$ with probability $1 - \mathsf{negl}(\lambda)$. We prove this in Lemma 8.6.
- **Underlying PCP consistency:** For any test $(Q_j, v_j) \in x^*$, if $\mathcal{Q}$ contains the indicator functions corresponding to $\{\Gamma_i\}_{i\in Q_j}$, the answers satisfy v_j. We prove this in Lemma 8.10.
- **Non-Signaling:** For any two ordered query sets $\mathcal{Q}_1$ and $\mathcal{Q}_2$, the answers to just the subset $S = \{i : \mathcal{Q}_{1,i} = \mathcal{Q}_{2,i}\}$ do not reveal whether $\mathcal{Q}_1$ or $\mathcal{Q}_2$ was queried. We prove this in Lemma 8.11.

We begin by proving an important helper lemma.

Lemma 8.3 (Hadamard answers on indicator vectors are consistent with local assignments)**.** *Let $j_1, \ldots, j_K \in [M]$. Consider an ordered query set $\mathcal{Q} = (q_1, \ldots, q_K) \subseteq \{0,1\}^M$ such that $q_i = e_{j_i}$ (i.e. the vector that is 1 at position j_i and 0 everywhere else). Let T be the subset of wires of $\widehat{\mathcal{C}}$ corresponding to $\{\Gamma_{j_i}\}_{i\in[K]}$ (where Γ is as defined in Step (2a) of the $\mathsf{SNARG}.\mathcal{P}$ algorithm). Then*

$$\Pr\left[\forall i \in [K], \Gamma_{j_i} = b_i \;\;:\;\; \begin{array}{c} (\{\Gamma_{j_i}\}_{i\in[K]}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T, \mathcal{Q}), \\ \forall i \in [K], c_i \leftarrow \mathsf{LHE.Dec}(\mathsf{sk}_i, \mathsf{ans}_i) \end{array}\right] \geq 1 - \mathsf{negl}(\lambda),$$

where we use the convention that $\mathsf{LHE.Dec}(\mathsf{sk}, 0) = 0$.

Proof. Fix any $v \in [K]$ and consider Γ_{j_v}. Recall $q_v = e_{j_v}$. In the following, we use the convention that $\mathsf{LHE.Dec}(\mathsf{sk}, 0) = 0$. Also, we refer to wire values $\alpha_{i,j}$ and $\beta_{i,j}$ as defined in Step (2c) of the $\mathsf{SNARG}.\mathcal{P}$ algorithm.

Claim 8.4. *For all $j \in [M]$, let $T_j = T \cup \{\alpha_{v,j}\}$. For $j \neq j_v$, we have:*

$$\Pr\left[\ \mathsf{LHE.Dec}(\mathsf{sk}_v, \alpha_{v,j}) = 0 \ \ : \ \ (\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\alpha_{v,j}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j, \mathcal{Q})\ \right] \geq 1 - \mathsf{negl}(\lambda)$$

and for $j = j_v$:

$$\Pr\left[\ \mathsf{LHE.Dec}(\mathsf{sk}_v, \alpha_{v,j_v}) = \Gamma_{j_v} \ \ : \ \ (\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\alpha_{v,j_v}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j, \mathcal{Q})\ \right] \geq 1 - \mathsf{negl}(\lambda).$$

Proof. Let S_j denote the set of wires containing $\mathsf{ct}_{v,j}$, and the wires that are used to compute $\alpha_{v,j}$ from Γ_j and $\mathsf{ct}_{v,j}$. Clearly, $|S_i| \leq \mathsf{poly}(\lambda)$. By local consistency of $\mathsf{LocalGen}^{\mathcal{A}}$, if

$$(\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\alpha_{v,j}\} \boxed{\cup (\sigma_i)_{i \in S_j}}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j \boxed{\cup S_j}, \mathcal{Q}),$$

it must be the case that

$$\alpha_{v,j} = \begin{cases} \mathsf{ct}_{v,j} & \text{if } \Gamma_j = 1 \\ 0 & \text{if } \Gamma_j = 0. \end{cases}$$

by its definition in $\widehat{\mathcal{C}}[x^*, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{ct}}]$. In particular, if $j \neq j_v$, since $\mathsf{ct}_{v,j}$ is an encryption of $q_{v,j} = 0$, by correctness of LHE, we have that $\mathsf{LHE.Dec}(\mathsf{sk}_v, \alpha_{v,j}) = 0$. By non-signaling of $\mathsf{LocalGen}^{\mathcal{A}}$, the same must hold when

$$(\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\alpha_{v,j}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j, \mathcal{Q}),$$

since a distinguisher $\mathcal{D}$ can compute $\mathsf{LHE.Dec}(\mathsf{sk}_v, \alpha_{v,j})$ given just sk_v (which is included in aux) and $\alpha_{v,j}$ (even without the wire values of S_j). This proves the first claim.

If $j = j_v$, then $\mathsf{ct}_{v,j}$ is an encryption of $q_{v,j_v} = 1$. Now, by a similar argument, we can argue via local consistency and non-signaling that $\mathsf{Dec}(\mathsf{sk}_v, \alpha_{v,j}) = \Gamma_j$. This proves the second claim. ☐

Claim 8.5. *For all $j \in [M]$, let $T_j = T \cup \{\beta_{v,j}\}$. There exists a polynomial μ such that the following holds. For all $j < j_v$,*

$$\Pr\left[\ \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}) = 0 \ \ : \ \ (\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\beta_{v,j}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j, \mathcal{Q}),\ \right] \geq 1 - j \cdot \mu(\lambda),$$

and for $j \geq j_v$, we have

$$\Pr\left[\ \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}) = \Gamma_{j_v} \ \ : \ \ (\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\beta_{v,j}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j, \mathcal{Q})\ \right] \geq 1 - j \cdot \mu(\lambda).$$

Proof. We proceed by induction. First consider $j = 1$. Let $T_1' = T_1 \cup \{\alpha_{v,1}\} = T \cup \{\beta_{v,1}\} \cup \{\alpha_{v,1}\}$. By local consistency of $\mathsf{LocalGen}^{\mathcal{A}}$, we have that

$$(\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\beta_{v,1}\} \cup \{\alpha_{v,1}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_1', \mathcal{Q})$$

satisfies $\beta_{v,1} = \alpha_{v,1}$ except with negligible probability. Moreover, by Claim 8.4, we have that with probability $1 - \mu'(\lambda)$ for some negligible function μ',

- if $j_v > 1$ then $\mathsf{LHE.Dec}(\mathsf{sk}_v, \alpha_{v,j}) = \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}) = 0$
- if $j_v = 1$ then $\mathsf{LHE.Dec}(\mathsf{sk}_v, \alpha_{v,j}) = \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}) = \Gamma_{j_v}$.

By non-signaling, the decryption of $\beta_{v,j}$ should not change even if we do not extract $\{\alpha_{v,1}\}$, and only extract

$$(\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\beta_{v,1}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_1, \mathcal{Q}).$$

Therefore, if we assume NS indistinguishability holds with advantage at most $\mu'(\lambda)$, then there exists $\mu = 2\mu'$ such that the claim holds for $j = 1$.

Now, suppose the claim holds for $j-1$. Let S_j denote the subcircuit used to compute $\beta_{v,j}$ from $\beta_{v,j-1}$ and $\alpha_{v,j}$. Let $T'_j = T_j \cup \{\beta_{v,j-1}, \alpha_{v,j}\} \cup S_j$.

By local consistency of $\mathsf{LocalGen}^{\mathcal{A}}$, we have that whenever

$$(\{\Gamma_{j_i}\}_{i\in[K]} \cup \{\beta_{v,j}\} \cup \{\beta_{v,j-1}, \alpha_{v,j}\} \cup S_j, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T'_j, \mathcal{Q}),$$

we must have that $\beta_{v,j} = \mathsf{LHE.Add}(\mathsf{pk}_v, \beta_{v,j-1}, \alpha_{v,j})$.

We have three cases to consider:

Case 1: $j < j_v$. If $j < j_v$, by the inductive hypothesis and non-signaling, we have that $\mathsf{LHE.Dec}(\beta_{v,j-1}) \neq 0$ with probability at most $(j-1)\cdot\mu(\lambda) + \mu'(\lambda)$. Similarly, by Claim 8.4 and non-signaling, we have that $\mathsf{LHE.Dec}(\alpha_{v,j}) \neq 0$ with probability at most $\mu_1(\lambda)$. Therefore, by a union bound, (and the correctness of LHE.Add) we have that $\mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}) \neq 0$ with probability at most $(j-1)\cdot\mu(\lambda) + 2\mu'(\lambda) \leq j\cdot\mu(\lambda)$.

Case 2: $j > j_v$. If $j-1 \geq j_v$, by the inductive hypothesis and non-signaling, we have that $\mathsf{LHE.Dec}(\beta_{v,j-1}) \neq \Gamma_{j_v}$ with probability at most $(j-1)\cdot\mu(\lambda)+\mu'(\lambda)$. Similarly, by Claim 8.4 and non-signaling, we have that $\mathsf{LHE.Dec}(\alpha_{v,j)}) \neq 0$ with probability at most $\mu_1(\lambda)$. Therefore, by a union bound (and the correctness of LHE.Add), we have that $\mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}) \neq \Gamma_{j_v}$ with probability at most $(j-1)\cdot\mu(\lambda) + 2\mu'(\lambda) \leq j\cdot\mu(\lambda)$.

Case 3: $j = j_v$. If $j = j_v$, by the inductive hypothesis and non-signaling, we have that $\mathsf{LHE.Dec}(\beta_{v,j-1}) \neq 0$ with probability at most $(j-1)\cdot\mu(\lambda)+\mu'(\lambda)$. Similarly, by Claim 8.4 and non-signaling, we have that $\mathsf{LHE.Dec}(\alpha_{v,j)}) \neq \Gamma_{j_v}$ with probability at most $\mu_1(\lambda)$. Therefore, by a union bound (and the correctness of LHE.Add), we have that $\mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}) \neq \Gamma_{j_v}$ with probability at most $(j-1)\cdot\mu(\lambda) + 2\mu'(\lambda) \leq j\cdot\mu(\lambda)$.

This completes the proof. □

Consider

$$\{\Gamma_{j_i}\}_{i\in[K]} \boxed{\cup\{\beta_{i,M}\}_{i\in[K]}}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T, \mathcal{Q}).$$

By local consistency of $\mathsf{LocalGen}^{\mathcal{A}}$, note that $\mathsf{ans}_i = \beta_{i,M}$. Moreover, by Claim 8.5 and non-signaling, each $\beta_{i,M}$ satisfies that $\mathsf{LHE.Dec}(\mathsf{sk}_i, \beta_{i,M}) = \Gamma_{j_i}$. In particular, this means that $\mathsf{LHE.Dec}(\mathsf{sk}_i, \mathsf{ans}_i) = \Gamma_{j_i}$. Therefore, by a union bound over all $i \in [K]$ and invoking non-signaling once more, we complete the proof. □

Lemma 8.6 (Answers are linearly consistent.). *Consider a query set $\mathcal{Q} = (q_1, \ldots, q_K \in \{0,1\}^M)$, and suppose q_u, q_v, q_w for $u, v, w \in [K]$ satisfies $q_u \oplus q_v \oplus q_w = 0$. Let $\mathcal{D}$ be the following distribution:*

- *Sample* $(b_1, \ldots, b_k) \leftarrow \mathsf{nsPCP\text{-}answers}(\mathcal{Q})$.
- *Output* $b_u \oplus b_v \oplus b_w$.

Then, $\mathcal{D}$ outputs 0 with probability $1 - \mathsf{negl}(\lambda)$.

Proof. We do this by first proving the following claim.

Claim 8.7. *Fix arbitrary $j \in [M]$. Let $T_j^\alpha = \{\alpha_{u,j}, \alpha_{v,j}, \alpha_{w,j}\}$. Then*

$$\Pr\left[c_u + c_v + c_w \in \{0, 2\} \quad : \quad \begin{array}{c} (\{\alpha_{u,j}, \alpha_{v,j}, \alpha_{w,j}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j^\alpha, \mathcal{Q}) \\ c_u = \mathsf{LHE.Dec}(\mathsf{sk}_u, \alpha_{u,j}), \\ c_v = \mathsf{LHE.Dec}(\mathsf{sk}_v, \alpha_{v,j}), \\ c_w = \mathsf{LHE.Dec}(\mathsf{sk}_w, \alpha_{w,j}), \end{array} \right]$$
$$\geq 1 - \mathsf{negl}(\lambda).$$

Proof. Consider the subcircuit of wires S which are used to compute $\alpha_{u,j}, \alpha_{v,j}, \alpha_{w_j}$ from $\mathsf{ct}_{u,j}, \mathsf{ct}_{v,j}, \mathsf{ct}_{w,j}$ and Γ_j. It is easy to see that this subcircuit has size $\mathsf{poly}(\lambda)$. By local consistency, we have that whenever:

$$(\{\alpha_{u,j}, \alpha_{v,j}, \alpha_{w,j}\}, \boxed{(\sigma_\omega)_{\omega \in S}}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j^\alpha \cup S, \mathcal{Q}),$$

we must have that for each $i \in \{u, v, w\}$,

$$\alpha_{i,j} = \begin{cases} \mathsf{ct}_{i,j} & \text{if } \Gamma_j = 1, \\ 0 & \text{if } \Gamma_j = 0. \end{cases}$$

By construction, if $\Gamma_j = 0$, all $c_i = \mathsf{LHE.Dec}(\mathsf{sk}_i, 0) = 0$ and $c_u + c_v + c_w = 0$. If $\Gamma_j = 1$, then $c_i = \mathsf{LHE.Dec}(\mathsf{sk}_i, \mathsf{ct}_{i,j}) = q_{i,j}$, and once again $c_u + c_v + c_w = q_{u,j} + q_{v,j} + q_{w,j} \in \{0, 2\}$. By $\mathsf{negl}(\lambda)$-non-signaling, we therefore have that the same holds if

$$(\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j^\alpha, \mathcal{Q}),$$

completing the proof. □

Now, using the above, we can prove the following claim.

Claim 8.8. *For $j \in [M]$, let $T_j^\beta = \{\beta_{u,j}, \beta_{v,j}, \beta_{w,j}\}$. There exists a negligible function μ such that the following holds.*

$$\Pr\left[c_u + c_v + c_w \in \{0, 2, \ldots, 2j\} \quad : \quad \begin{array}{c} (\{\beta_{u,j}, \beta_{v,j}, \beta_{w,j}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j^\beta, \mathcal{Q}) \\ c_u = \mathsf{LHE.Dec}(\mathsf{sk}_u, \beta_{u,j}), \\ c_v = \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}), \\ c_w = \mathsf{LHE.Dec}(\mathsf{sk}_w, \beta_{w,j}), \end{array} \right]$$
$$\geq 1 - j \cdot \mu(\lambda).$$

Proof. Let μ' be a negligible function bounding the non-signaling indistinguishability as in Claim 8.2, and the bound from Claim 8.7. We will prove this via induction.

We first consider $j = 1$. Let T be the set containing $\alpha_{i,1}$ for $i \in \{u, v, w\}$, and all the wires S used to compute $\beta_{i,1}$ from $\alpha_{i,1}$. Recall that $|T| \leq \ell$. By local consistency, we have that

$$(\{\beta_{u,1}, \beta_{v,1}, \beta_{w,1}\}, \boxed{\{\alpha_{u,1}, \alpha_{v,1}, \alpha_{w,1}\}, (\sigma_\omega)_{\omega \in S}}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_1^\beta \cup T, \mathcal{Q})$$

satisfies that $\beta_{i,1} = \alpha_{i,1}$ (unless it outputs $\perp$). In particular, $\mathsf{LHE.Dec}(\mathsf{sk}_i, \beta_{i,1}) = \mathsf{LHE.Dec}(\mathsf{sk}_i, \alpha_{i,1})$, therefore $c_u + c_v + c_w \in \{0, 2\}$ by Claim 8.7. By non-signaling, we can deduce that:

$$\Pr\left[c_u + c_v + c_w \in \{0, 2\} \;:\; \begin{array}{r} (\{\beta_{u,1}, \beta_{v,1}, \beta_{w,1}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_1^\beta, \mathcal{Q}) \\ c_u = \mathsf{LHE.Dec}(\mathsf{sk}_u, \beta_{u,1}), \\ c_v = \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,1}), \\ c_w = \mathsf{LHE.Dec}(\mathsf{sk}_w, \beta_{w,1}), \end{array} \right] \geq 1 - O(1) \cdot \mu'(\lambda).$$

Now, suppose the claim holds for $j - 1$. For $i \in \{u, v, w\}$, let T contain $\beta_{i,j-1}, \alpha_{i,j-1}$, and all the wires S used to compute $\beta_{i,j}$ from them. By local consistency, we have that:

$$\begin{aligned} &(\{\beta_{u,j}, \beta_{v,j}, \beta_{w,j}\}, \boxed{\{\alpha_{u,j}, \alpha_{v,j}, \alpha_{w,j}\}, (\sigma_\omega)_{\omega \in S}}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}, \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \\ &\qquad \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j^\beta \cup T, \mathcal{Q}) \end{aligned}$$

satisfies that $\beta_{i,j} = \mathsf{LHE.Add}(\mathsf{pk}_i, \beta_{i,j-1}, \alpha_{i,j-1})$. By correctness of $\mathsf{LHE.Add}$, we have that

$$\mathsf{LHE.Dec}(\mathsf{sk}_i, \beta_{i,j}) = \mathsf{LHE.Dec}(\mathsf{sk}_i, \beta_{i,j-1}) + \mathsf{LHE.Dec}(\mathsf{sk}_i, \alpha_{i,j-1}) \in \{0, 2, \ldots, 2(j-1) + 2\}.$$

Therefore, by Claim 8.7 and the inductive hypothesis and non-signaling, we have that

$$\Pr\left[c_u + c_v + c_w \in \{0, 2, \ldots, 2(j-1)+2\} \;:\; \begin{array}{c} (\{\beta_{u,j}, \beta_{v,j}, \beta_{w,j}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \\ \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_j^\beta \cup S, \mathcal{Q}), \\ c_u = \mathsf{LHE.Dec}(\mathsf{sk}_u, \beta_{u,j}), \\ c_v = \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,j}), \\ c_w = \mathsf{LHE.Dec}(\mathsf{sk}_w, \beta_{w,j}), \end{array} \right] \geq 1 - (j-1) \cdot \mu(\lambda) + O(1) \cdot \mu'(\lambda).$$

By choosing $\mu = D \cdot \mu'$ for large enough $D \in \mathbb{N}$, we prove the claim. $\square$

Now, consider T_M^β, and consider

$$(\{\beta_{u,M}, \beta_{v,M}, \beta_{w,M}\}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, T_M^\beta, \mathcal{Q}).$$

By local consistency of $\mathsf{LocalGen}^{\mathcal{A}}$, we have that $\mathsf{ans}_u = \beta_{u,M}$, $\mathsf{ans}_v = \beta_{v,M}$, $\mathsf{ans}_w = \beta_{w,M}$. By Claim 8.8, we know that

$$\mathsf{LHE.Dec}(\mathsf{sk}_u, \beta_{u,M}) + \mathsf{LHE.Dec}(\mathsf{sk}_v, \beta_{v,M}) + \mathsf{LHE.Dec}(\mathsf{sk}_w, \beta_{w,M}) \neq 0 \mod 2$$

with probability at most $\mathsf{negl}(\lambda)$. Therefore, by non-signaling, and a union bound, we have that the claim indeed holds. $\square$

A simple modification of the above proof can also be used to prove the following theorem.

Lemma 8.9 (Consistency among answers)**.** *Consider a query set $\mathcal{Q} = (q_1, \ldots, q_K \in \{0,1\}^M)$, and suppose q_u, q_v for $u, v \in [K]$ satisfies $q_u \oplus q_v = 0$. Let $\mathcal{D}$ be the following distribution:*

- *Sample* $(b_1, \ldots, b_k) \leftarrow \mathsf{nsPCP\text{-}answers}(\mathcal{Q})$.
- *Output* $b_u \oplus b_v$.

Then, $\mathcal{D}$ outputs 0 with probability $1 - \mathsf{negl}(\lambda)$.

Proof (sketch). This proof follows via a simple modification to the proof of Lemma 8.6. The main difference is that we prove Claim 8.7 and Claim 8.8 with $c_u + c_v = 0$ rather than $c_u + c_v + c_w = 0$. $\square$

Lemma 8.10 (Answers pass PCP tests)**.** *Let $(Q_j = (s_1, \ldots, s_k), v_j)$ be an arbitrary test in x^*. For $i \in [k]$, let e_i be the indicator vector corresponding to:*

$$e_{i,r} = \begin{cases} 1 & \textit{if } r = s_i, \\ 0 & \textit{otherwise.} \end{cases}$$

In other words, $\langle e_i, \Gamma \rangle = \Gamma_{s_i}$. Consider $\mathcal{Q} = (q_1, \ldots, q_K)$ such that $q_{j_i} = e_i$. Let $\mathcal{D}$ be the following distribution:

- *Sample* $(b_1, b_2, b_3, \ldots, b_K) \leftarrow \mathsf{nsPCP\text{-}answers}(\mathcal{Q})$.
- *Output* $v_j(b_{j_1}, \ldots, b_{j_k})$.

Then, $\mathcal{D}$ outputs 0 with probability $1 - \mathsf{negl}(\lambda)$.

Proof. Let S be the set of wires $\{\Gamma_{s_1}, \ldots, \Gamma_{s_k}\}$. Let T be the set of wires involved in the check that $v_j(\Gamma_{Q_j}) = 1$, the wires out, and the path from ρ_j to the root in the binary tree computing out in Step (2b) of the SNARG.$\mathcal{P}$ algorithm above. By construction, $|T| \leq P + \log t$. Consider

$$(\{\Gamma_{s_i}\}_{i \in [k]} \cup (\sigma_\omega)_{\omega \in T}, (\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \overrightarrow{\mathsf{sk}}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}^{\mathcal{A}}(x^*, S \cup T, \mathcal{Q}).$$

By the success probability (which implies that $\mathsf{out} = 1$) and local consistency of $\mathsf{LocalGen}^{\mathcal{A}}$, we have that all values along the path from out to ρ_j (including ρ_j) must be 1, and thus $v_j(\Gamma_{s_1}, \ldots, \Gamma_{s_k}) = 0$ except with negligible probability (unless $\mathsf{LocalGen}^{\mathcal{A}}$ outputs 1). Now, let $b_i = \mathsf{LHE.Dec}(\mathsf{sk}_i, \mathsf{ans}_i)$. By invoking Lemma 8.3, we have that $b_{j_i} = \Gamma_{s_i}$ except with negligible probability. In particular, we therefore have that $v_j(b_{j_1}, \ldots, b_{j_k}) = 0$ except with negligible probability. By non-signaling of $\mathsf{LocalGen}^{\mathcal{A}}$, this completes the proof. $\square$

Lemma 8.11 (Non-Signaling on Tuples)**.** *Consider two ordered query sets $\mathcal{Q}_0$ and $\mathcal{Q}_1$. Let $S = \{i : Q_{0,i} = Q_{1,i}\}$. Consider $\mathcal{D}_\beta$ which works as follows:*

- *Query* $(b_1, \ldots, b_K) \leftarrow \mathsf{nsPCP\text{-}answers}(\mathcal{Q}_\beta)$.
- *Output* $\{b_i\}_{i \in S}$.

We claim that the outputs of $\mathcal{D}_0$ and $\mathcal{D}_1$ have statistical distance $\mathsf{negl}(\lambda)$.

Proof. We will first show that the two distributions are computationally indistinguishable. Then, we can invoke Fact 3.4 and the fact that $K = O(1)$ to show that the two distributions have statistical distance $\mathsf{negl}(\lambda)$.

In order to compute $\{b_i\}_{i \in S}$, note that nsPCP-answers only needs the keys $\{\mathsf{sk}_i\}_{i \in S}$. Consider TDGen' which outputs $\mathsf{td}' = \{\mathsf{sk}_i\}_{i \in S}$ rather than $\overrightarrow{\mathsf{sk}}$, and consider $\mathsf{LocalGen}'$ which is $\mathsf{LocalGen}^{\mathcal{A}}$ instantiated with TDGen' rather than TDGen. It is easy to see that the following is a functionally equivalent way to generate $\mathcal{D}_b$.

- Call $\{(\mathsf{out}, \overrightarrow{\mathsf{ans}}), \overrightarrow{\mathsf{pk}}, \{\mathsf{sk}_i\}_{i \in S}, \overrightarrow{\mathsf{ct}}) \leftarrow \mathsf{LocalGen}'(x^*, \varnothing, \mathcal{Q})$. (If $\mathsf{LocalGen}'$ fails, output $\perp$).
- For $i \in S$, compute $b_i = \mathsf{LHE.Dec}(\mathsf{sk}_i, \mathsf{ans}_i) \pmod 2$.
- Output $\{b_i\}_{i \in S}$.

Therefore, to show that $\mathcal{D}_0$ and $\mathcal{D}_1$ are computationally indistinguishable, it suffices to show that the outputs of $\mathsf{SNARG.TDGen}'(1^\lambda, 1^n, 1^m, \mathcal{Q}_0)$ and $\mathsf{SNARG.TDGen}'(1^\lambda, 1^n, 1^m, \mathcal{Q}_1)$. Indeed, the only difference in the TDGen' algorithm in the two cases appears in Step (2). In this step, $\mathsf{ct}_j = \mathsf{Enc}(\mathsf{pk}_j, \mathcal{Q}_{b,j})$ for $j \in [K]$, and the sampling procedure differs only on $j \notin S$ (all other ciphertexts are sampled identically). While $\overrightarrow{\mathsf{pk}}$ is public, $\{\mathsf{sk}_j\}_{j \notin S}$ remain hidden, allowing us to invoke the semantic security of the LHE scheme to complete the proof. ☐

8.2.3 To a Sherali-Adams Pseudoexpectation

Finally, we consider an algorithm which takes as input (unordered) sets $\mathcal{Q}$ of size at most $K' \leq K$.

nsPCP-answers-unordered($\mathcal{Q}$):

- Parse $\mathcal{Q} = \{q_1, \ldots, q_{K'}\}$ where the the queries are ordered via some canonical ordering.
- Let T be a tuple $(q_1, \ldots, q_{K'}, 0, \ldots, 0)$ of size K which is padded with $K - K'$ zeros.
- Call $(b_1, \ldots, b_K) \leftarrow$ nsPCP-answers(T).
- Output $\{b_1, \ldots, b_{K'}\}$.

First, note that the answers provided by this algorithm inherit the first two consistency properties we proved about nsPCP-answers in Section 8.2.2, i.e., that they pass the Hadamard-encoded nsPCP tests. In other words, Lemmas 8.6 and 8.10 are true even if $(b_1, \ldots, b_K) \leftarrow$ nsPCP-answers-unordered($\mathcal{Q}$), because nsPCP-answers-unordered($\mathcal{Q}$) only outputs the answers from nsPCP-answers on $q \in \mathcal{Q}$.

We additionally prove a non-signaling property of the answers provided by this algorithm, following the proof of [JKLV24, Lemma 4.3].

Lemma 8.12 (Non-Signaling on Sets). *Consider two query sets $\mathcal{Q}_0$ and $\mathcal{Q}_1$. Consider $\mathcal{D}_\beta$ which works as follows:*

- *Query* $(b_1, \ldots, b_{K'}) \leftarrow$ nsPCP-answers-unordered($\mathcal{Q}_\beta$).
- *Output* $\{b_i\}_{q_i \in \mathcal{Q}_0 \cap \mathcal{Q}_1}$.

We claim that the outputs of $\mathcal{D}_0$ and $\mathcal{D}_1$ have statistical distance $\mathsf{negl}(\lambda)$.

Proof. We will first show that the two distributions are computationally indistinguishable. Then, we can invoke Fact 3.4 and the fact that $K = O(1)$ to show that the two distributions have statistical distance $\mathsf{negl}(\lambda)$.

Consider the tuples $T^{(\beta)} = (q_1^{(\beta)}, \ldots, q_{K'}^{(\beta)}, 0, \ldots, 0)$ which are queried to $\mathsf{nsPCP\text{-}answers}$ by $\mathsf{nsPCP\text{-}answers\text{-}unordered}(\mathcal{Q}_\beta)$. Let $\mathcal{Q}_0 \cap \mathcal{Q}_1 = (q_1^*, \ldots, q_L^*)$ ordered in the canonical ordering used by $\mathsf{nsPCP\text{-}answers\text{-}unordered}$.

For $i \in [L]$, let $s_i^{(\beta)}$ be the index of q_i^* in $T^{(\beta)}$. Let π be a permutation on $[K']$ which maps $s_i^{(0)}$ to $s_i^{(1)}$ for all $i \in [L]$. We can write π as the composition of at most K' transpositions, and we denote the composition of the first j of these transpositions as π_j, so $\pi_{K'} = \pi$. For $j \in [K']$, define the distribution $\mathcal{D}_j$ which samples $(b_1, \ldots, b_K) \leftarrow \mathsf{nsPCP\text{-}answers}(q^{(0)}_{\pi_j^{-1}(1)}, \ldots, q^{(0)}_{\pi_j^{-1}(K')}, 0, \ldots, 0)$ and outputs $\{b_{\pi_j(s_i^{(0)})}\}_{i \in [L]}$. Using Lemma 8.9 and Lemma 8.11, we can see by induction that for all $j \in [K']$, $\mathcal{D}_j$ is computationally indistinguishable from $\mathcal{D}_0$. We conclude that $\mathcal{D}_{K'}$, is computationally indistinguishable from $\mathcal{D}_0$.

Consider what $\mathcal{D}_{K'}$ samples and outputs. Since $\pi_{K'} = \pi$, and π maps $s_i^{(0)}$ to $s_i^{(1)}$ for all $i \in [L]$, we have that $\mathcal{D}_{K'}$ queries $q_i^* = q^{(0)}_{s_i^{(0)}} = q^{(1)}_{s_i^{(1)}}$ at position $\pi(s_i^{(0)}) = s_i^{(1)}$, and then outputs $\{b_{s_i^{(1)}}\}_{i \in [L]}$. Thus by Lemma 8.11, we have that $\mathcal{D}_{K'}$ and $\mathcal{D}_1$ are computationally indistinguishable.

$\square$

Let $\mathcal{D}_{\mathcal{Q}} = \{\mathcal{D}_{\mathcal{Q}}\}_{\mathcal{Q} \subseteq \{0,1\}^M, |\mathcal{Q}| \le K}$ denote the non-signaling strategy (i.e., family of local distributions) obtained by calling $\mathsf{nsPCP\text{-}answers\text{-}unordered}(\mathcal{Q})$ for $\mathcal{Q} \subseteq \{0,1\}^M$ of size at most K. Lemma 8.12 shows that the strategy $\mathcal{D}_{\mathcal{Q}}$ satisfies (K, δ)-non-signaling (as in Definition 3.5, where $\delta = \mathsf{negl}(\lambda)$. By applying Theorem 3.6, there exists a $(K, 0)$-non-signaling strategy $\mathcal{D}'_{\mathcal{Q}} = \{\mathcal{D}'_{\mathcal{Q}}\}_{\mathcal{Q} \subseteq \{0,1\}^M, |\mathcal{Q}| \le K}$ such that

$$\mathcal{D}_{\mathcal{Q}} \overset{\eta}{\equiv} \mathcal{D}'_{\mathcal{Q}}$$

for $\eta \le 2^K \delta + 2^{2K} \delta = \mathsf{negl}(\lambda)$ (recall $K = O(1)$).

It is easy to see that the strategy $\mathcal{D}'_{\mathcal{Q}}$ still passes the Hadamard-encoded nsPCP tests: if there existed a query set $\mathcal{Q}^*$ such that $\mathcal{D}'_{\mathcal{Q}}$ fails some test with non-negligible probability, it must have non-negligible distance from $\mathcal{D}_{\mathcal{Q}}$ on the query set $\mathcal{Q}^*$.

By Claim 3.18, there exists a pseudoexpectation $\widetilde{\mathbb{E}}$ over variables $\{x_i\}_{i \in \{0,1\}^M}$, which we interpret as Hadamard-extended variables $\{x_S, y_T\}_{S \subseteq [m], T \subseteq [\rho(m)]}$, such that for any Hadamard-encoded nsPCP test polynomial $p(x, y)$ we have that $\widetilde{\mathbb{E}}[p(x, y)^2]$ is the probability that $\mathcal{D}'_{\mathcal{Q}}$ fails the test, which is negligible. This completes the soundness analysis.

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A Hash-and-BARG construction and analysis

In this section, we recap the hash-and-BARG construction from [CJJ22, KVZ21, BBK+23] and argue that it is a quasi-argument satisfying the properties in Definition 4.10. The construction relies on two main ingredients:

- A hash tree family $\mathsf{HT} = (\mathsf{HT.Gen}, \mathsf{HT.Hash}, \mathsf{HT.Open}, \mathsf{HT.Verify})$.
- A somewhere-extractable BARG family for index-relations, given by $\mathsf{seBARG} = (\mathsf{seBARG.Gen}, \mathsf{seBARG}.\mathcal{P}, \mathsf{seBARG}.\mathcal{V}, \mathsf{seBARG.TDGen}, \mathsf{seBARG.Extract})$.

We now show how to use these to construct a quasi-argument scheme. For convenience, we will define two additional algorithms $\mathsf{HB.TDGen}$ and $\mathsf{HB.LocalTupGen}^{\mathcal{A}}$ which will simplify the security analysis.

- $\mathsf{HB.Setup}(1^\lambda, 1^s, 1^\ell)$:
  1. Sample $\mathsf{hk} \leftarrow \mathsf{HT.Gen}(1^\lambda)$.
  2. Let $p = \mathsf{poly}(\lambda)$ be the size of the relation defined in Step (3) of the $\mathsf{HB}.\mathcal{P}$ algorithm below. Sample ℓ independent seBARG common reference strings $\mathsf{crs}_1, \ldots, \mathsf{crs}_\ell \leftarrow \mathsf{seBARG.Gen}(1^\lambda, s, 1^p)$.
  3. Output $(\mathsf{hk}, \{\mathsf{crs}_i\}_{i\in[\ell]})$.
- $\mathsf{HB}.\mathcal{P}(\mathsf{crs}, C, w)$:
  1. Evaluate the circuit $y \leftarrow C(w)$, and let τ correspond to the wire assigments of the evaluations.
  2. Compute $h \leftarrow \mathsf{Hash}(\mathsf{hk}, \tau)$.
  3. Consider the following relation indexed by the hash h, hash key hk, circuit C and output y denoted by
  $$R_{\mathsf{hk},h,C,y}(i, (b_1, b_2, b_3, o_1, o_2, o_3))$$
  which outputs 1 only if the following conditions hold:
     - Let the indices j_1, j_2 denote the labels of the input wires, and j_3 denotes the label of the output wire of the ith gate g_i.
     - **Valid openings:** For all $k \in [3]$, $\mathsf{HT.Verify}(\mathsf{hk}, h, j_k, b_k, o_k) = 1$.
     - **Consistency of the gate:** The gate $g_i(b_1, b_2) = b_3$.

- **Circuit output consistency:** If wire j_3 corresponds to the mth output wire of C, additionally check that $y_m = b_3$.

4. For each gate g_i of C, compute $w_i = (b_1, b_2, b_3, o_1, o_2, o_3)$, where
   - Set indices (j_1, j_2, j_3) corresponding to the indices of the wires of g_i.
   - Set bit $b_k \leftarrow \tau_{j_k}$.
   - Set opening $o_k \leftarrow \mathsf{HT.Open}(\mathsf{hk}, \tau, j_k)$.
5. For $k = 1, \ldots, \ell$: Compute $\pi_k = \mathsf{seBARG}.\mathcal{P}(\mathsf{crs}, R_{\mathsf{hk},h,C,y}, \{w_i\}_{i \in [s]})$.
6. Output $y, \pi := (h, \{\pi_k\}_{k \in [\ell]})$.

- $\mathsf{HB}.\mathcal{V}(\mathsf{crs}, C, \pi, y)$:
  1. Parse $\mathsf{crs} = (\mathsf{hk}, \{\mathsf{crs}_i\}_{i \in [\ell]}\}$, and $\pi = (h, \{\pi_k\}_{k \in [\ell]})$.
  2. Construct relation $R_{\mathsf{hk},h,C,y}$.
  3. For $i \in [\ell]$, run $\mathsf{seBARG}.\mathcal{V}(\mathsf{crs}_i, R_{\mathsf{hk},h,C,y}, \pi_i)$. Accept if all seBARG proofs are accepted, reject otherwise.

- $\mathsf{HB.TDGen}(1^\lambda, 1^\ell, C, T)$:
  1. Let $T = (t_1, \ldots, t_\ell)$ be an ordered tuple of wires.
  2. Pick indices $i_1, \ldots, i_\ell \in [s]$ such that the wire
     - if t_k is an input wire, pick an arbitrary index i_k such that t_k is incident to gate g_{i_k},
     - else, pick i_k so that t_k is the output wire of g_{i_k}.
  3. Sample $\mathsf{hk} \leftarrow \mathsf{HT.Gen}(1^\lambda)$.
  4. For $k \in [\ell]$, sample $(\mathsf{crs}_k, \mathsf{td}_k,) \leftarrow \mathsf{seBARG.TDGen}(1^\lambda, s, 1^p, i_k)$. Sample the remaining CRS via $\mathsf{seBARG.Gen}(1^\lambda, s, 1^p)$.
  5. Set $\mathsf{crs} := (\mathsf{hk}, \{\mathsf{crs}_k\}_{k \in [\ell]})$ and $\mathsf{td} = (\{\mathsf{td}_1\}_{k \in [\ell]})$.
  6. Output $(\mathsf{crs}, \mathsf{td})$.

- $\mathsf{HB.LocalTupGen}^{\mathcal{A}}(C, T)$:
  1. Let $T = (t_1, \ldots, t_\ell)$ be an ordered tuple.
  2. Call $(\mathsf{crs}, \mathsf{td}) \leftarrow \mathsf{HB.TDGen}(1^\lambda, 1^\ell, C, T)$.
  3. Invoke $\mathcal{A}$ on crs, and receive (y, π, aux).
  4. Parse $\pi = (h, \{\pi_k\}_{k \in [\ell]})$.
  5. If $\mathsf{HB}.\mathcal{V}(\mathsf{crs}, C, \pi, y) = 0$, output $(\perp, \perp, \perp)$.
  6. For $k \in [\ell]$, extract $w_k \leftarrow \mathsf{seBARG.Extract}(\mathsf{td}_k, \pi_k)$, and set b_k to be the bit of w_k corresponding to wire t_k.
  7. Output $(y, \{b_k\}_{k \in [\ell]}, \mathsf{aux})$.

- $\mathsf{HB.LocalGen}^{\mathcal{A}}(C, S)$:

1. Let $S = \{s_1, \ldots, s_{\ell'}\}$ be a set of size $\ell' \leq \ell$, ordered in some canonical ordering.
2. Let $T = (s_1, \ldots, s_{\ell'}, 0, \ldots, 0)$ be a tuple of size ℓ (i.e. padded with enoughs zeros).
3. Call $(y, \{b_k\}_{k \in [\ell]}, \mathsf{aux}) \leftarrow \mathsf{HB.LocalTupGen}^{\mathcal{A}}(C, T)$.
4. Output $(y, \{b_k\}_{k \in [\ell']}, \mathsf{aux})$.

Theorem A.1. *The tuple of algorithms* $(\mathsf{HB.Setup}, \mathsf{HB}.\mathcal{P}, \mathsf{HB}.\mathcal{V})$ *along with the ppt oracle machine* $\mathsf{HB.LocalGen}^{\mathcal{A}}$ *is a publicly verifiable quasi-argument system as defined in Definition 4.10.*

Since this proof has been implicit in many works (see e.g. [JKLV24]), we only give a sketch of the proof.

Proof sketch. We prove that the above algorithm satisfies all the properties in Definition 4.10.

Completeness and efficiency. The completeness and efficiency of the scheme follow directly from the opening correctness of HT and the completeness of the seBARG scheme.

Correct distribution. It suffices to show that an adversary $\mathcal{A}$ cannot distinguish between a crs generated from $\mathsf{HB.Gen}(1^\lambda, 1^\ell)$ and $\mathsf{HB.TDGen}(1^\lambda, 1^\ell, C, \varnothing)$ are indistinguishable (because the correctness experiment is identical otherwise). Indistinguishability follows directly from seBARG index hiding.

Local consistency. It suffices to consider local consistency (which extends in a natural way) of $\mathsf{HB.LocalTupGen}^{\mathcal{A}}$ (since the $\mathsf{LocalTupGen}$ consistency implies consistency of $\mathsf{LocalGen}$). Consider some $(y, \{b_k\}_{k \in [\ell']}, \mathsf{aux}) \leftarrow \mathsf{HB.LocalTupGen}^{\mathcal{A}}(C, T)$ such that $y \neq \bot$.

Case 1: Suppose T contains some wire out_j corresponding to an output wire, and let b_i denote the corresponding output value. We claim that $b_i = y_j$. Indeed, recall that $\mathsf{crs} \leftarrow \mathsf{HB.TDGen}(1^\lambda, 1^\ell, C, T)$ is generated so that $\mathsf{crs}_i \leftarrow \mathsf{seBARG.TDGen}(1^\lambda, s, 1^p, j_i)$, where the index j_i corresponds to the output gate incident to out_j. By somewhere soundness (implied by somewhere extractability) of this seBARG, we know that any accepting BARG proof under crs_i must indeed satisfies "circuit output consistency", i.e. that $b_i = y_j$.

Case 2: Suppose T contains t_i, t_j, t_k such that they are all incident to same gate g such that $t_i \leftarrow g(t_j, t_k)$. Now, consider witnesses w_i, w_j, w_k as defined in Step (6) of the $\mathsf{HB.LocalTupGen}$ algorithm above. Parse the following:

- Parse $w_i = (b_i, \beta_2, \beta_3, o_1, o_2, o_3)$.
- Parse w_j to obtain the bit and opening b_j, o_2' corresponding to wire t_j.
- Parse w_k to obtain the bit b_k, o_3' corresponding to wire t_k.

By the collision resistance of HT, we have that with probability $1 - \mathsf{negl}(\lambda)$, $\beta_2 = b_j$ and $\beta_3 = b_3$ (otherwise one can use o_2, o_2' or o_3, o_3' to violate the collision resistance). Therefore, by somewhere soundness, w_i is guaranteed to satisfy that $b = g(\beta_2, \beta_3) = g(b_j, b_k)$.

This completes the proof of consistency.

Wire consistency. Before we show non-signaling, we first show an additional property satisfied by $\mathsf{LocalTupGen}$. Consider a tuple $T = (t_1, \dots, t_\ell)$ such that $t_i = t_j$. We claim that

$$\Pr\left[\begin{array}{c}\sigma_i = \sigma_j \\ \vee\, y = \bot\end{array} : (y, (\sigma_i)_{i\in[\ell]}, \mathsf{aux}) \leftarrow \mathsf{QA.LocalTupGen}^{\mathcal{A}}(C, T)\right] \geq 1 - \mathsf{negl}(\lambda)$$

This follows from collision-resistance of the hash function exactly as argued in the case of local consistency.

Non-signaling. We will first show a different non-signaling guarantee on *ordered tuples*. Let $T, T' \subseteq [s]^\ell$. Write $T = (t_1, \dots, t_\ell)$ and $T' = (t'_1, \dots, t'_\ell)$. Let $U \subseteq [\ell]$ be the set of all indices i such that $t_i = t'_i$. We then claim that:

$$\Big|\Pr\left[\ \mathcal{D}(y, (\sigma_i)_{i\in U}, \mathsf{aux}) = 1 \ :\ (y, (\sigma_i)_{i\in[\ell]}, \mathsf{aux}) \leftarrow \mathsf{QA.LocalGen}^{\mathcal{A}}(C, T)\ \right]$$
$$- \Pr\left[\ \mathcal{D}(y', (\sigma'_i)_{i\in U}, \mathsf{aux}') = 1 \ :\ (y', (\sigma'_i)_{i\in[\ell]}, \mathsf{aux}') \leftarrow \mathsf{QA.LocalGen}^{\mathcal{A}}(C, T')\ \right]\Big| \leq \mathsf{negl}(\lambda).$$

It suffices to show that the output of $\mathsf{HB.TDGen}(1^\lambda, 1^\ell, C, T)$ in the and $\mathsf{HB.TDGen}(1^\lambda, 1^\ell, C, T')$ are indistinguishable in the presence of $\{\mathsf{td}_i\}_{i\in U}$, where td_i is as defined in Step (4) in $\mathsf{HB.TDGen}$ (these trapdoors are sufficient to extract $(\sigma_i)_{i\in U}$ in Step (6) of $\mathsf{HB.LocalTupGen}$). The indistinguishability of the CRS follows directly from seBARG index hiding. This completes the proof.

We can then use standard techniques (see for example, [JKLV24, Lemma 4.3], and Lemma 8.12) to show that non-signaling and wire consistency of $\mathsf{LocalTupGen}^{\mathcal{A}}$ implies the non-signaling property of $\mathsf{LocalGen}^{\mathcal{A}}$ as defined in Definition 4.10. □

B Proof sketch of Theorem 1.2

In this section, we outline a proof sketch of Theorem 1.2. The starting point of this proof is the observation that the universal SNARG scheme of [JKLM25] is secure if there exists any two-message laconic argument with an extended Frege proof of correctness that only satisfies a notion of *weak soundness*.

Definition B.1. A (uniform)[14] two message laconic argument scheme consists of PPT algorithms $(\mathcal{V} = (\mathcal{V}_1, \mathcal{V}_2), \mathcal{P})$ for a language $\mathcal{L}$ with the following syntax.

- $\mathcal{V}_1(1^\lambda, x)$ generates a message$(\mathsf{pk}, \mathsf{vk}) \leftarrow \mathcal{V}_1(1^\lambda, x)$.
- $\mathcal{P}$ takes as input a pair $(x, w) \in \mathcal{R}_{\mathcal{L}}$ and pk and outputs a message message $\pi = \mathcal{P}(\mathsf{pk}, x, w)$.
- $\mathcal{V}_2$ takes as input a pair (vk, x, π) and outputs a verdict bit b.

We say that the argument system is publicly verifiable if $\mathsf{vk} = \varnothing$, and privately verifiable otherwise.

The scheme is additionally required to satisfy the efficiency and completeness as defined below.

[14] In the work of [JKLM25], the algorithm $\mathcal{V}$ is additionally given some aux_x depending on the instance x. In this work, it suffices to consider uniform algorithms.

Completeness: For all $\lambda \in \mathbb{N}$, all polynomials $n(\lambda)$ and witness w such that $C_{\mathcal{L}}(x, w) = 1$ (where $C_{\mathcal{L}}$ is the NP verification circuit for $\mathcal{L}$),

$$\Pr\left[\mathcal{V}_2(\mathsf{vk}, x, \pi) = 1 \ : \ \begin{array}{c} (\mathsf{pk}, \mathsf{vk}) \leftarrow \mathcal{V}(1^\lambda, x), \\ \pi \leftarrow \mathcal{P}(\mathsf{pk}, x, w) \end{array} \right] = 1.$$

We say that the scheme has am Extended Frege ($\mathcal{EF}$) proof of completeness if there exists a polynomial p such that the claim that "For every w, if $C_x(w) = 1$ then $\mathcal{V}_2(\mathsf{vk}, x, \mathcal{P}(\mathsf{pk}, x, w)) = 1$" has a $\mathcal{EF}$ proof of size $p(\lambda, n)$. For full details on $\mathcal{EF}$, we refer the reader to [JKLM25].

Soundness: For $x \notin \mathcal{L}$, we have that for sufficiently large λ, and all polynomial sized adversaries $\mathcal{A}$,

$$\Pr\left[\mathcal{V}_2(\mathsf{vk}, x, \pi) = 1 \ : \ \begin{array}{c} (\mathsf{pk}, \mathsf{vk}) \leftarrow \mathcal{V}(1^\lambda, x), \\ \pi \leftarrow \mathcal{A}^*(\mathsf{pk}) \end{array} \right] \leq \mathsf{negl}(\lambda).$$

Efficiency: In the completeness experiment, $|\pi| = \mathsf{poly}(\lambda, \log n)$.

We now define a weaker soundness notion for two-message laconic arguments that will be sufficient for the universal SNARG construction of JKLM.

Definition B.2 (Weak soundness of a two-message laconic argument.)**.** Let $(\mathcal{V}, \mathcal{P})$ be a two message laconic argument system. Let $\mathcal{D}_{\lambda,n}$ denote the distribution of randomness used by $\mathcal{V}_1(1^\lambda, \cdot)$ on instances of length n, and let $\mathcal{U}_{\lambda,n}$ denote the support of $\mathcal{D}_{\lambda,n}$. We say that a two-message laconic argument is weakly sound if for all sufficiently large λ and polynomial-sized adversary $\mathcal{A}$, there exists some distribution $\mathcal{D}'$ over $\mathcal{U}_{\lambda,n}$ such that

$$\Pr\left[\mathcal{V}_2(\mathsf{vk}, x, \pi) = 1 \ : \ \begin{array}{c} r \leftarrow \mathcal{D}' \\ (\mathsf{pk}, \mathsf{vk}) \leftarrow \mathcal{V}(1^\lambda, x; r), \\ \pi \leftarrow \mathcal{A}^*(\mathsf{pk}) \end{array} \right] \leq 1 - \frac{1}{\mathsf{poly}(\lambda)}.$$

We emphasize that the order of quantifiers states that we can pick $\mathcal{D}'$ after picking the adversary.

Theorem B.3. *Assume the polynomial hardness of* LWE*. Fix an* NP *language* $\mathcal{L}$ *for which there exists a weakly sound two message laconic argument with a proof of correctness. Then, there exists a* SNARG *for* $\mathcal{L}$[15] *with a CRS of size* $\mathsf{poly}(\lambda, n)$ *and proof of size* $\mathsf{poly}(\lambda, \log n)$*.*

The proof follows nearly identically to the proof of [JKLM25, Theorem 4.1].

We now recall the designated-verifier SNARG of Kalai, Raz and Rothblum [KRR14], instantiated via a PCP with k = $O(1)$ locality. We assume that that even if the PCP has exponential size, the ith bit can be computed efficiently. We will denote this by $\mathsf{PCP}.\mathcal{P}(x, w, i)$.

- $\mathsf{KRR.Gen}(1^\lambda, 1^n; \mathsf{sd}_{\mathcal{Q},0}, \mathsf{sd}_{\mathcal{Q},1}, \mathsf{sd}_{\mathsf{sk}} = \{\mathsf{sd}_{\mathsf{sk},b,i}\}_{b\in\{0,1\},i\in[k]}, \mathsf{sd}_{\mathsf{ct}} = \{\mathsf{sd}_{\mathsf{ct},b,i}\}_{b\in\{0,1\},i\in[k]})$:
  - For $b \in \{0,1\}$, use $\mathsf{sd}_{\mathcal{Q},b}$ to generate PCP queries $q_{b,1}, \ldots, q_{b,k} \leftarrow \mathsf{PCP}.\mathcal{Q}(1^n)$.
  - For $b \in \{0,1\}, i \in [k]$, generate FHE keys $\mathsf{sk}_{b,i} \leftarrow \mathsf{FHE.Gen}(1^\lambda; \mathsf{sd}_{\mathsf{sk},b,i})$.
  - For $b \in \{0,1\}, i \in [k]$, generate encryptions $\mathsf{ct}_{b,i} \leftarrow \mathsf{FHE.Enc}(\mathsf{sk}_{b,i}, q_{b,i}; \mathsf{sd}_{\mathsf{ct},b,i})$.
  - Output $\mathsf{td} = (\{\mathsf{sk}_{b,i}\}_{b\in\{0,1\},i\in[k]}, \{q_{b,k}\}_{b\in\{0,1\},i\in[k]})$ and $\mathsf{crs} = \{\mathsf{ct}_{b,i}\}_{b\in\{0,1\},i\in[k]}$.

[15] One can be more explicit about the size of the CRS and proof as a function of the keys and proof sizes of the underlying laconic argument system. We gloss over this detail in this work, but refer the reader to [JKLM25] for a more refined analysis.

- $\mathsf{KRR}.\mathcal{P}(\mathsf{crs}, x, w)$:
  - Parse $\mathsf{crs} = \{\mathsf{ct}_{b,i}\}_{b\in\{0,1\},i\in[k]}$.
  - Homomorphically evaluate $\mathsf{PCP}.\mathcal{P}(x, w, \cdot)$ on $\mathsf{ct}_{b,i}$ to obtain $\mathsf{ans}_{b,i}$.
  - Output $\pi = \{\mathsf{ans}_{b,i}\}_{b\in\{0,1\},i\in[k]}$.
- $\mathsf{KRR}.\mathcal{V}(\mathsf{crs}, \mathsf{td}, x, \pi)$:
  - Parse $\mathsf{td} = (\{\mathsf{sk}_{b,i}\}_{b\in\{0,1\},i\in[k]}, \{q_{b,k}\}_{b\in\{0,1\},i\in[k]})$ and $\pi = \{\mathsf{ans}_{b,i}\}_{b\in\{0,1\},i\in[k]}$.
  - For $b \in \{0, 1\}, i \in [k]$, compute $a_{i,b} \leftarrow \mathsf{FHE.Dec}(\mathsf{sk}_{b,i}, \mathsf{ans}_{b,i})$.
  - Check for $b \in \{0, 1\}$, $\mathsf{PCP}.\mathcal{V}((q_{b,1}, \ldots, q_{b,k}), (a_{b,1}, \ldots, a_b, k)) = 1$, accept.
  - For every $q_{0,i} = q_{1,i'}$, verify that $\mathsf{ans}_{0,i} = \mathsf{ans}_{1,i'}$.
  - If any of the checks fail, output 0. Else, output 1.

As mentioned in [JKLM25], a designated-verifier SNARG is in fact a two-message laconic argument defined by $\mathcal{V}_1 = \mathsf{KRR.Gen}$, $\mathcal{V}_2 = \mathsf{KRR}.\mathcal{V}$, $\mathcal{P} = \mathsf{KRR}.\mathcal{P}$, and $\mathsf{pk} = \mathsf{crs}$ and $\mathsf{vk} = \mathsf{td}$. It is easy to see that if both the FHE scheme and the PCP scheme have a proof of correctness, then the resulting designated-verifier SNARG also has a proof of correctness.

Lemma B.4. *Suppose that the above designated-verifier SNARG scheme does not satisfy weak soundness. Then, there exists a non-signaling strategy for the* PCP *which violates weak non-signaling soundness.*

Proof (sketch). Suppose that there exists an adversary $\mathcal{A}$ such that for every distribution $\mathcal{D}'$ on the randomness used in Gen, we have that

$$\Pr\left[\mathsf{KRR}.\mathcal{V}(\mathsf{vk}, x, \pi) = 1 \ : \ \begin{array}{c} r \leftarrow \mathcal{D}' \\ (\mathsf{crs}, \mathsf{td}) \leftarrow \mathsf{KRR.Gen}(1^\lambda, x; r), \\ \pi \leftarrow \mathcal{A}^*(\mathsf{crs}) \end{array}\right] \geq 1 - \mathsf{negl}(\lambda).$$

We now consider the following algorithm that generates PCP answers.

$\mathsf{nsPCP\text{-}answers}(Q_0, Q_1)$:

1. Choose $\mathsf{sd}_{\mathcal{Q},b}$ so that $Q_b = \mathsf{PCP}.\mathcal{Q}(1^n, \mathsf{sd}_{\mathcal{Q}})$.
2. Sample random $\mathsf{sd}_{\mathsf{sk}}$ and $\mathsf{sd}_{\mathsf{ct}}$.
3. Let $(\mathsf{td}, \mathsf{crs}) \leftarrow \mathsf{KRR.Gen}(1^\lambda, 1^n; \mathsf{sd}_{\mathcal{Q},0}, \mathsf{sd}_{\mathcal{Q},1}, \mathsf{sd}_{\mathsf{sk}}, \mathsf{sd}_{\mathsf{ct}})$.
4. Query $\pi \leftarrow \mathcal{A}(\mathsf{crs})$.
5. Parse $\mathsf{td} = (\{\mathsf{sk}_{b,i}\}_{b\in\{0,1\},i\in[k]}, \{q_{b,k}\}_{b\in\{0,1\},i\in[k]})$ and $\pi = \{\mathsf{ans}_{b,i}\}_{b\in\{0,1\},i\in[k]}$.
6. For $b \in \{0, 1\}, i \in [k]$, compute $a_{i,b} \leftarrow \mathsf{FHE.Dec}(\mathsf{sk}_{b,i}, \mathsf{ans}_{b,i})$.
7. Output $\{a_{b,i}\}_{b\in\{0,1\},i\in[k]}$.

By the guarantee that $\mathcal{A}$ violates weak soundness, we have that with probability $1-\mathsf{negl}(\lambda)$, $\{a_i\}_{i\in[k]}$ passes $\mathsf{PCP}.\mathcal{V}(Q, \{a_i\}_{i\in[k]}$. Moreover, one can show via FHE semantic security that the answers satisfy non-signaling on tuples, i.e. for tuples Q and Q' of length $2k$, the marginal distribution of answers on $\{j : Q_j = Q'_j\}$ is non-signaling. Moreover, recall that if Q_0 and Q_1 contain a common query, they are consistent with each other with probability $1 - \mathsf{negl}(\lambda)$.

Now, we construct non-signaling distribution on sets of queries of size at most k.

> $\mathsf{nsPCP\text{-}answers\text{-}unordered}(Q_0)$:
>
> - Without loss of generality, suppose that Q_0 is in some canonical ordering.
> - Choose a random Q_1 in the support of $\mathsf{PCP}.\mathcal{Q}(1^n)$.
> - Call $\{a_{b,i}\}_{b\in\{0,1\},i\in[k]} \leftarrow \mathsf{nsPCP\text{-}answers}(Q_0, Q_1)$.
> - Output $\{a_{0,i}\}_{i\in[k]}$.

The non-signaling property of the above construction follows from a standard argument to go from non-signaling tuples satisfying consistency to non-signaling sets (see for example, Lemma 8.12).

Therefore, the above algorithm gives a family of distributions $\{\mathcal{D}_S\}_S$ which is computationally non-signaling and satisfies the PCP tests. By invoking Theorem 3.6 to obtain a perfectly non-signaling family $\{\mathcal{D}'_S\}_S$ where each $\mathcal{D}_S$ and $\mathcal{D}'_S$ have statistical distrance $\mathsf{negl}(\lambda)$. Therefore, the resulting family also satisfies the PCP tests with probability $1 - \mathsf{negl}(\lambda)$. This demonstrates a non-signaling strategy for the underlying PCP which violates weak non-signaling soundness. □